Operator-system Lorentzian pre-length spaces and the strong-form Krein–Mondino–Sämann bridge: cross-KMS-state thermal-time stack and strict super-additivity

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Abstract

We introduce the category $\mathbf{OSLPLS}_{\mathrm{cov}}$ of *operator-system Lorentzian pre-length spaces* (OSLPLS), the non-commutative extension of Mondino–Sämann's covered Lorentzian pre-length space framework [34] obtained by replacing the underlying set X with the state space $\mathcal{S}(\mathcal{M})$ of an operator-system topography $\mathcal{M}$, the time separation ℓ with the modular-flow time separation $\ell_\omega^{\mathrm{OS}}$ on a faithful basepoint state ω, and the cover by sub-operator-systems $(\mathcal{M}_k)_{k\in\mathbb{N}}$. The commutative Mondino–Sämann category $\mathbf{LorPLG}_{\mathrm{cov}}$ embeds in $\mathbf{OSLPLS}_{\mathrm{cov}}$ as a faithful sub-category via $X \mapsto C(X)$, $\ell \mapsto L_\ell$, $o \mapsto \delta_o$.

Main result. The bridge functor

$$W^{\mathrm{flip}} : \mathbf{KreinMetaMet}_{\mathrm{pp}}^{\mathrm{flip,full}} \longrightarrow \mathbf{OSLPLS}_{\mathrm{cov}}$$

extends the Paper 48 [60] K^+-weak-form bridge $\mathbf{W}$ to the full $M \neq 0$ Paper 46 [58] Appendix B enlarged Krein substrate. The *Bridge Theorem 6.4′-Q1′* (Theorem 7.5) is stated on four properties, the metric-dependent ones of which are descoped (see the Status note: the properties quantifying over the upstream Paper 45/46 metric quantity are open pending repair; the thermal-time-stack content is state-level and survives): (B1′) structural correspondence inherited from the OSLPLS axiom transport (Theorem 3.1); (B2′) reverse triangle inequality with *strict* super-additivity on three-orbit triples, established via the **Connes–Rovelli thermal-time stack across distinct KMS states**, a construction not previously published; (B3′) pre-compactness via the Paper 44 [56] propagation-number bound prop ≤ 4 on the enlarged substrate. The fourth property, (B4′) convergence transport, is *descoped*: it would follow from Paper 46's strong-form propinquity bound, but that seminorm is degenerate (Paper 45/46), so the metric-level convergence leg is not established.

The thermal-time stack. The substantive new mathematics of this paper sits in Section 6. Three structural ingredients combine: (i) the *orbit-KMS lemma* (Lemma 6.1): every modular orbit on the bridge image carries an intrinsic effective KMS state inherited from the Bisognano–Wichmann vacuum via GNS restriction; (ii) the *Connes (1973) cocycle Radon–Nikodym derivative* [9] $u_t^{\sigma_i\sigma_j} = [D\omega^{\sigma_j} : D\omega^{\sigma_i}]_t$ intertwines distinct modular flows on shared sub-operator-systems and provides a cross-orbit thermal-time bridge; (iii) the *triple-intersection cocycle identity* (TICI, Theorem 6.2): $u_t^{\sigma_1\sigma_3} = u_t^{\sigma_1\sigma_2} \cdot \sigma_t^{\omega^{\sigma_1}}(u_t^{\sigma_2\sigma_3})$ on the triple overlap, implicit in Connes 1973 (C4) but not previously isolated as a stack-consistency property. Strict super-additivity then follows from the max-divergence chain inequality (elementary from the $D_{\max}$ definition; max-divergence due to Datta [28]) applied to the cocycle entropy production: the detoured worldline through an intermediate KMS state pays *two* max-divergence cocycle costs $\Delta_{\max}S^{1\to2} + \Delta_{\max}S^{2\to3}$ while the direct worldline pays only $\Delta_{\max}S^{1\to3}$, with the Datta chain inequality forcing $\Delta_{\max}S^{1\to3} \leq \Delta_{\max}S^{1\to2} + \Delta_{\max}S^{2\to3}$,

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strictly on a generic set of triples (Theorem 6.3; pairwise distinctness is necessary but not sufficient). The Umegaki relative entropy itself does not satisfy this chain inequality in general (see §1.4); the max-divergence is the chain-stable surrogate.

Physical reading: twin paradox as quantum information. The strict super-additivity of ℓ^{OS} on off-orbit triples is the operator-algebraic dual of the special-relativistic twin paradox. Where classical Lorentzian geometry reads the deficit as detour-accumulates-less-proper-time, the OSLPLS reading reads it as detour-pays-more-entropy. The max-divergence chain inequality (Datta 2009 [28]) supplies the strict-inequality content. To the author's knowledge this is the first explicit identification of an information-theoretic underpinning for the reverse triangle inequality of synthetic Lorentzian geometry.

Numerical panel. The bit-exact Paper 45 [57] joint propinquity panel $\Lambda^L(n_{\max}, N_t) \in \{2.0746, 1.6101, 1.3223\}$ at $(n_{\max}, N_t) \in \{(2,3), (3,5), (4,7)\}$ transports verbatim through the strong-form bridge. An illustrative Umegaki relative-entropy panel shows substantially positive deficits ($\sim$ 67, 69, 81 nats); the load-bearing $D_{\max}$ chain is verified at 96/96 cells (`tests/test_wh7_b3_phase3_sprint2.py`). Riemannian-limit recovery at $N_t = 1$ is bit-exact (Frobenius residual = 0.0 in float64). Propagation number prop = 2 is verified at $(n_{\max}, N_t) = (2,3)$ (Paper 46 Appendix B envelope-aware refinement predicts prop ≤ 4).

Honest scope. The bridge is established at the **strict-strong-form OSLPLS level** on the full $M \neq 0$ enlarged substrate, at finite cutoff. The closed-form expression for the cocycle entropy production $\Delta S^{\sigma_i \to \sigma_j}(t)$ as a function of OSLPLS state-space coordinates is named open. The *genuine non-commutative Mondino–Sämann pre-length space* concept (Q2′, Section 11) remains the named NCG-research frontier, beyond sprint scale; OSLPLS *replaces* Mondino–Sämann rather than generalising it on the synthetic side. Cross-manifold $(\mathcal{T}_{S^3} \otimes \mathcal{T}_{\mathrm{Hardy}(S^5)})$ and inner-factor calibration data are orthogonal to the present strong-form bridge.

Place in the math.OA standalone series. Eleventh in the GeoVac math.OA arc (siblings: Papers 29 [49], 32 [50], 38 [51], 39 [52], 40 [53], 42 [54], 43 [55], 44 [56], 45 [57], 46 [58], 47 [59], 48 [60]). Builds on Paper 48 (K$^+$-weak-form bridge) as the direct predecessor: Paper 49 lifts the bridge to the strict-strong-form by extending the target category from $\mathbf{LorPLG}_{\mathrm{cov}}$ to $\mathbf{OSLPLS}_{\mathrm{cov}}$ and the source substrate from M-diagonal to full $M \neq 0$ enlarged. The K$^+$-weak-form bridge of Paper 48 sits in W^{flip} as a structurally degenerate sub-case (Decomposition O Case (iii) empty); the strict-strong-form content of Paper 49 activates Case (iii) non-trivially.

Status and scope (June 2026)

Three issues are established upstream of this paper: (i) the $\mathcal{K}^+$-restricted compressed-Dirac Lipschitz seminorm underlying Λ^L vanishes identically on the truncated operator system — Krein-self-adjointness of the anti-Hermitian spatial part $i\,D_{\mathrm{GV}} \otimes I$ forces $\{J_L, D_{\mathrm{GV}} \otimes I\} = 0$, so the $\mathcal{K}^+$ compression annihilates the spatial Dirac, and the momentum-diagonal temporal multipliers commute with ∂_t (verified bit-exact at $(n_{\max}, N_t) \in \{(2,3), (3,5)\}$); (ii) the "Latrémolière Thm 5.5 / Def. 3.4 / Def. 3.5" references cited for the tunneling-pair bound do not exist in the cited source — the device actually used is van Suijlekom's state-space Gromov–Hausdorff framework; (iii) the constant $2/(n_{\max}+1)$ is the maximum of the Plancherel mass distribution, not a cb-norm of any map used in the proofs (the smoothing map is unital completely positive, with cb-norm 1).

Consequently the bit-exact Λ^L panel inheritance (§ 8.1) and the K$^+$ framing inherited from Papers 45 and 48 carry the upstream descope: the transported values are evaluations of the same closed-form rate formula, not measurements of a quantum-metric distance. The bridge-theorem properties of Theorem 7.5 that quantify over that degenerate metric quantity — in particular the convergence transport (B4′) and the metric-ball reading of the pre-compactness property (B3′) — are *descoped pending the upstream repair.* What survives unaffected: the cocycle entropy-production (max-divergence) deficit computations of § 8.2 as state-level numerics (modular theory and the Datta chain inequality, not the Lipschitz seminorm); the TICI cocycle algebra (Theorem 6.2) and the strict super-additivity theorem (Theorem 6.3) as statements about KMS states; and the OSLPLS categorical design. The enlarged substrate of chirality-flipping generators with $\{J, M\} = 0$ (Paper 46 Appendix B) is where the Lipschitz seminorms are genuinely non-zero, making it the natural starting substrate for the repair. The Lorentzian quantum-metric convergence question is an open named research target.

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1 Introduction

1.1 The Q1′ question: strong-form Krein–MS bridge

The K$^+$-weak-form Krein–Mondino–Sämann bridge of Paper 48 [60] constructs a contravariant functor

$$\mathbf{W} : \mathbf{KreinMetaMet}_{\mathrm{pp}} \longrightarrow \mathbf{LorPLG}_{\mathrm{cov}}$$

that sends a Krein-pointed proper quantum metric space $(\mathcal{A}^K, L^K, \mathcal{M}^L, \omega^L_W)$ on the natural Lorentzian operator-system substrate of Paper 44 [56] to a covered Lorentzian pre-length space $(\hat{\mathcal{M}}^L, \ell^L, \hat{\omega}^L_W, \hat{\mathcal{U}}^L)$ via Gelfand spectrum, modular-flow time separation, Bisognano–Wichmann vacuum basepoint, and admissible-scaling truncated cover. The four Bridge Theorem properties (B1)–(B4) hold at K$^+$-weak-form rigor, with the reverse triangle inequality (B2) closing through a *Decomposition O* analysis on causal triples: Case (i) on-orbit holds with equality from Paper 42 [54] four-witness modular flow additivity, Cases (ii), (iv) are trivial via $\ell^L = -\infty$ on causally-unrelated pairs, and the substantive Case (iii) (three characters on three distinct modular orbits with all pairs causally related) is *structurally empty* on the K$^+$-weak-form substrate because the M-diagonal topography forces orbit-label preservation under modular flow.

Paper 48 §8.1 names this gap as the open question *Q1′*: extend **W** to the strict-strong-form by enlarging the source substrate (Paper 46 [58] Appendix B chirality-flip generators at full $M \neq 0$) so that Decomposition O Case (iii) becomes non-empty, and prove the reverse triangle with *strict* super-additivity on three-orbit triples. The Paper 48 honest scope acknowledges Q1′ as a substantial NCG-mathematics follow-on, beyond sprint scale: activating Case (iii) requires the chirality-flipping generators to carry non-trivial m_j-rotation content (i.e. to be multiplication operators on the spinor bundle with non-zero magnetic quantum number), which breaks Abelianness of the topography and takes the construction outside the Gelfand–Naimark setting.

Paper 49 closes Q1′ at the OSLPLS / state level (the metric-level convergence leg, B4′, is descoped).

1.2 Three structural moves

The closure rests on three structural moves.

(i) Replace Mondino–Sämann with OSLPLS as the synthetic-side target. The Q1′-Light diagnostic (§2.3) showed that enlarging the topography to the full $M \neq 0$ chirality-flip substrate breaks Abelianness: for two chirality-flip generators $M^{\mathrm{flip}}_{N_1,L_1,M_1} = \mathrm{diag}(W^{N_1L_1M_1}, -W^{N_1L_1M_1})$ and $M^{\mathrm{flip}}_{N_2,L_2,M_2}$, the commutator reduces to the spatial-multiplier commutator $[W^{N_1L_1M_1}, W^{N_2L_2M_2}]$ on each chirality sector, which is non-trivial whenever $M_1, M_2 \neq 0$ (Wigner-rotation content on the m_j labels). The Gelfand spectrum ceases to apply directly. We replace the target category $\mathbf{LorPLG}_{\mathrm{cov}}$ by the category $\mathbf{OSLPLS}_{\mathrm{cov}}$ of operator-system Lorentzian pre-length spaces, in which the underlying set is the state space $\mathcal{S}(\mathcal{M})$ of the operator-system topography $\mathcal{M}$, the time separation is generated by the Tomita–Takesaki modular flow on a faithful basepoint state ω, and the cover is by sub-operator-systems $(\mathcal{M}_k)_{k\in\mathbb{N}}$. The OSLPLS category contains Mondino–Sämann as a faithful sub-category via $X \mapsto C(X)$ (Theorem 3.7).

(ii) Identify W^{flip} on objects and morphisms. The bridge functor $W^{\mathrm{flip}} : \mathbf{KreinMetaMet}^{\mathrm{flip,full}}_{\mathrm{pp}} \to \mathbf{OSLPLS}_{\mathrm{cov}}$ acts on objects by direct operator-system transport: source $(\mathcal{A}^K, L^K, \mathcal{M}^{L,\mathrm{flip}}_{\mathrm{full}}, \omega^L_W) \mapsto (\mathcal{A}^K, L^K, \mathcal{M}^{L,\mathrm{flip}}_{\mathrm{full}}, \omega^L_W|_{\mathcal{M}^{L,\mathrm{flip}}_{\mathrm{full}}}, \mathcal{U}^{L,\mathrm{flip}}_{\mathrm{OS}})$ with no Gelfand-spectrum step. On morphisms W^{flip} acts by Connes–van Suijlekom contravariant UCP convention [12]: a topographic Krein M-isometry on the enlarged substrate corresponds at the algebra level to a $*$-homomorphism viewed in the OSLPLS contravariant direction (Theorem 4.7). The construction is forced by the axiom transport analysis (Theorem 3.1): once one requires the bridge target category to (a) contain commutative MS as a faithful sub-category and (b) admit the Krein-side enlarged-substrate image, there is essentially no design freedom in the OSLPLS object or morphism class.

(iii) Connes–Rovelli thermal-time stack across distinct KMS states. The substantive new mathematics is the cross-KMS-state extension of the Connes–Rovelli thermal-time hypothesis (§6). The hypothesis of Connes–Rovelli [11] identifies modular flow at a fixed KMS state as the physical time evolution. The Bratteli–Robinson account [5], Theorem 5.3.10, together with Connes' cocycle Radon–Nikodym derivative [9], provides the cross-KMS-state intertwiner. We isolate the *triple-intersection cocycle identity* (TICI, Theorem 6.2) implicit in Connes 1973 (C4) but not previously named as a stack-consistency property, and use it together with Uhlmann's relative-entropy monotonicity [29, 44] to prove strict super-additivity of ℓ^{OS} on three-orbit triples (Theorem 6.3).

1.3 The headline: twin paradox as quantum information

Substituting the cocycle entropy production into the OSLPLS reverse triangle inequality and using Lindblad's data-processing inequality yields the operator-algebraic dual of the special-relativistic twin paradox. Consider three states $\omega_x \in \mathcal{O}^{\sigma_1}, \omega_y \in \mathcal{O}^{\sigma_2}, \omega_z \in \mathcal{O}^{\sigma_3}$ on three distinct modular orbits with all three pairs causally related via cross-orbit cocycle transports. The

OSLPLS time separation decomposes as

$$\ell^{\mathrm{OS}}(\omega', \omega'') \;=\; t^{\sigma_i \to \sigma_j}_{\mathrm{therm}} \;-\; \Delta S^{\sigma_i \to \sigma_j},$$

where the cocycle entropy production $\Delta S^{\sigma_i \to \sigma_j} \geq 0$ is the irreversible work of cross-orbit transport. The detoured worldline $\omega_x \to \omega_y \to \omega_z$ pays *two* max-divergence cocycle costs $\Delta_{\max} S^{1\to 2} + \Delta_{\max} S^{2\to 3}$; the direct worldline $\omega_x \to \omega_z$ pays *one* cost $\Delta_{\max} S^{1\to 3}$. the max-divergence chain inequality (elementary from the $D_{\max}$ definition; [28]) $\Delta_{\max} S^{1\to 3} \leq \Delta_{\max} S^{1\to 2} + \Delta_{\max} S^{2\to 3}$ (strict generically) translates to

$$\ell^{\mathrm{OS}}(\omega_x, \omega_y) + \ell^{\mathrm{OS}}(\omega_y, \omega_z) \;<\; \ell^{\mathrm{OS}}(\omega_x, \omega_z),$$

which is the Mondino–Sämann reverse triangle inequality at strict inequality. The detoured worldline accumulates less proper time *because* the detour pays more entropy-production cost.

To the author's knowledge this is the first explicit identification of quantum relative-entropy monotonicity as the load-bearing structural ingredient for the reverse triangle inequality of synthetic Lorentzian geometry. The reverse triangle inequality is well established in classical Lorentzian geometry as a property of proper time along causal curves [20, 34]; the OSLPLS reading reveals that the same inequality at the operator- algebraic level rests on a fundamentally information-theoretic inequality (the data-processing inequality of Lindblad [29]).

1.4 Honest scope and place in the math.OA series

Paper 49 is the eleventh math.OA standalone in the GeoVac project, building on the K^+-weak-form Krein–MS bridge of Paper 48 [60] (the tenth) by extending the bridge to strict-strong-form on the full $M \neq 0$ enlarged substrate. The bridge is established at finite cutoff with admissible-scaling for the cover; the continuum (norm-resolvent) extension parallels Paper 47 [59] on the K^+-weak-form bridge and is a separate Phase-3+ target.

The honest scope statement (§11):

- Paper 49 closes Q1′ at the OSLPLS-target level (Option γ from the Q1′-Light diagnostic §2.3). The strict-strong-form Bridge Theorem 6.4′-Q1′ holds at theorem-grade rigor on the enlarged substrate for its structural and thermal-time-stack legs (B1′/B2′/B3′); the metric-level convergence leg B4′ is descoped.

- Paper 49 does *not* close Q2′: the genuine non-commutative Mondino–Sämann pre-length space concept built from scratch (Option δ). OSLPLS is a category that *contains* Mondino–Sämann via the embedding functor ι (Theorem 3.7); it is not itself an extension of the Mondino–Sämann *concept* of pre-length space. Q2′ remains the named NCG-research frontier (§11.1).

- A closed-form expression for the cocycle entropy production $\Delta S^{\sigma_i \to \sigma_j}(t)$ as a function of OSLPLS state-space coordinates is named open (§11.2). Theorem 6.3 establishes strict super-additivity qualitatively via the data-processing inequality but does not quantify the deficit. The numerical panel (§8) is an illustrative Umegaki relative-entropy cross-check (deficits $\sim$ 67, 69, 81 nats at three cells); the load-bearing $D_{\max}$ chain is verified at 96/96 cells, but the closed-form quantification remains open.

- Cross-manifold extensions ($\mathcal{T}_{S^3} \otimes \mathcal{T}_{\mathrm{Hardy}(S^5)}$, blocked at the NCG-framework level by the Paper 24 [48] §V four-layer Coulomb/HO asymmetry) and inner-factor calibration data (the W3-question of Paper 32 [50] §VIII.C) are orthogonal to the present strong-form bridge.

1.5 Roadmap of the paper

Section 2 recalls the necessary background: the Paper 48 K^+-weak-form Krein–MS bridge (§2.1), the Paper 46 Appendix B enlarged substrate (§2.2), Mondino–Sämann covered Lorentzian pre-length spaces (§2.4), and the Connes–Rovelli thermal-time hypothesis with Connes 1973 cocycle Radon–Nikodym machinery (§2.5). The Q1′-Light diagnostic findings establishing the $M = 0/M \neq 0$ bimodal-Abelianness bifurcation are recalled in §2.3.

Section 3 defines the OSLPLS category. The axiom-transport analysis (Theorem 3.1) walks through each Mondino–Sämann axiom and identifies its operator-system generalization, surfacing the substantive observation that the reverse triangle inequality is the only MS axiom whose *proof* uses commutativity in a load-bearing way. The OSLPLS object (Definition 3.2) and morphism (Definition 3.4) are then specified, with the modular-intertwining axiom (M5) shown to be automatic from basepoint-state pullback (M3) via the functoriality of Tomita–Takesaki (Lemma 3.5). The commutative Mondino–Sämann category embeds as a faithful sub-category (Theorem 3.7).

Section 4 constructs the Krein-side bridge candidate W^{flip} on objects and morphisms. The Phase-1 chirality-graded bridge functor $W^{\mathrm{flip},M=0}$ at the $M = 0$ enlarged sub-substrate is recalled as a structurally additive stepping stone (§4.1); W^{flip} at the full $M \neq 0$ enlarged substrate is then defined as the operator-system extension (Definition 4.3). All five OSLPLS object axioms verify on $W^{\mathrm{flip}}(\mathbb{X}^{K,\mathrm{flip},\mathrm{full}})$ (Theorem 4.4), with the substantive new content being the structural activation of Decomposition O Case (iii) via non-trivial modular flow on chirality-flip generators at $M \neq 0$.

Section 5 catalogues the substantive activation of off-orbit modular structure at full $M \neq 0$ and recalls why the Phase-1 Case A construction at $M = 0$ fails to activate it. Modular flow generated by $K^W_\alpha = J_{\mathrm{polar}}$ mixes m_j labels on the chirality-flip generators $M^{\mathrm{flip}}_{N,L,M}$ at $M \neq 0$, so different states on $\mathcal{S}(\mathcal{M}^{L,\mathrm{flip}}_{\mathrm{full}})$ can sit on genuinely different modular orbits with non-trivial ℓ^{OS} relations.

Section 6 is the substantive heart of the paper. We construct the Connes–Rovelli thermal-time stack across distinct KMS states in three steps: the orbit-KMS lemma (Lemma 6.1) gives the intrinsic effective KMS structure on each modular orbit; the Connes 1973 cocycle Radon–Nikodym derivative (§6.2) provides the cross-orbit intertwiner with its cocycle composition rule; the triple-intersection cocycle identity (TICI, Theorem 6.2) isolates the stack-consistency property at triple overlaps. We then prove strict super-additivity (Theorem 6.3) via Uhlmann's relative-entropy monotonicity, with the physical reading as the twin-paradox-as-quantum-information identification.

Section 7 states and proves the aggregate Bridge Theorem 6.4′-Q1′ on the OSLPLS-target (Theorem 7.5), combining: B1′ structural correspondence (mechanical from Theorem 3.1), B2′ strict super-additivity (substantive, from §6), B3′ pre-compactness inheritance via Paper 44 propagation-number bound with envelope-aware refinement (prop ≤ 4 on the enlarged substrate), and B4′ convergence transport from Paper 46's strong-form propinquity bound.

Section 8 reports the numerical verification panel: Lambda joint propinquity bit-exact at $\{(2, 3), (3, 5), (4, 7)\}$, max-divergence cocycle deficits substantially positive, Riemannian-limit recovery at $N_t = 1$ bit-exact, propagation number $= 2$ verified at the smallest cell.

Section 9 discusses the synthetic-Lorentzian implications: the bridge image of the GeoVac hemispheric wedge at full $M \neq 0$ is a candidate non-commutative Lorentzian pre-length space (in the OSLPLS sense), positioning the construction relative to the Mondino–Sämann–Sormani–Cavalletti synthetic-Lorentzian-GH community.

Section 11 catalogues the open questions: Q2′ non-commutative MS extension, closed-form cocycle entropy production deficit, higher-genus orbit topology, cross-KMS thermal time across modular intertwiners, G3 cross-manifold, G4 inner-factor calibration.

1.6 Related work

The Latrémolière lineage [13, 14, 21–25] provides the operator-algebraic side of the bridge. The Mondino–Sämann lineage [20, 33, 34] provides the synthetic side. Adjacent synthetic-Lorentzian convergence frameworks include Sormani–Vega null distance [40], Sakovich–Sormani timed Gromov–Hausdorff [38], Minguzzi–Suhr bounded Lorentzian metric spaces [32], Müller's Cauchy slabs [35], Allen–Burtscher metric GH [1], Che–Perales–Sormani timed Hausdorff [8], Ketterer's ℓ-convergence [18], Kubota's coarea inequality [19].

The Connes–van Suijlekom operator-system framework [12] provides the substrate for OSLPLS (UCP morphisms, propagation number, operator-system Lipschitz seminorms). The Marcolli–van Suijlekom gauge-network construction [30] is the structurally nearest published reading of finite-dimensional spectral triples as graph-localised structures, of which the GeoVac spectral triple is a member [50].

The Connes–Rovelli thermal-time hypothesis [11] for a single KMS state is the literature template for the construction in §6; Paper 49's contribution is the cross-KMS-state extension, isolating the triple-intersection cocycle identity as the load-bearing stack-consistency property and applying it via Uhlmann's inequality [29, 44, 47] to give strict super-additivity.

To the author's knowledge, OSLPLS as a category is genuinely new: no published "operator-system Lorentzian pre-length space" concept exists as of May 2026. The closest adjacent direction is the Lorentzian-NCG literature of [3, 15, 36, 41, 46], which builds Lorentzian- signature spectral triples (Krein-space construction); OSLPLS sits one level up, taking the operator-system substrate of a Lorentzian spectral triple (Paper 44 [56]) and extracting the synthetic-Lorentzian metric structure via Tomita–Takesaki modular flow.

2 Setup

2.1 The Paper 48 K$^+$-weak-form bridge

Paper 48 [60] establishes the Wick-rotation functor $\mathbf{W} : \mathbf{KreinMetaMet}_{\mathrm{pp}} \to \mathbf{LorPLG}_{\mathrm{cov}}$ on the natural Lorentzian operator-system substrate of Paper 44 [56]. The Krein-pointed proper quantum metric space is the 4-tuple

$$\mathbb{X}^K = (\mathcal{A}^K, L^K, \mathcal{M}^L, \omega_W^L),$$

with $\mathcal{A}^K$ the natural Lorentzian operator system on the compact-temporal Lorentzian Krein space $\mathcal{K}_{n_{\max},N_t,T}$ [56], L^K the operator-norm Lipschitz seminorm $L^K(a) = \|[D_L, a]\|_{\mathrm{op}}$ [58], $\mathcal{M}^L$ the M-diagonal Abelian topography

$$\mathcal{M}^L = \mathrm{span}_{\mathbb{C}}\big\{M^{\mathrm{spat}}_{N,L,0} \otimes I_{N_t} \,:\, N \le n_{\max},\ L < N\big\} \subseteq \mathcal{A}^K,$$

and $\omega_W^L = e^{-K_\alpha^W}/Z$ the Bisognano–Wichmann wedge KMS state with $K_\alpha^W = J^W_{\mathrm{polar}}$ of integer spectrum two_m$_j$ on the spinor bundle [54].

The Wick-rotation functor sends $\mathbb{X}^K$ to the covered LPLS

$$\mathbf{W}(\mathbb{X}^K) = (\hat{\mathcal{M}}^L, \ell^L, \hat{\omega}_W^L, \hat{\mathcal{U}}^L)$$

with $\hat{\mathcal{M}}^L = \mathrm{Spec}(\mathcal{M}^L)$ the Gelfand spectrum of the (Abelian) M-diagonal topography, $\ell^L = \kappa_g \cdot \tau_{\mathrm{mod}}^{\omega_W^L}$ the modular-flow time separation ($-\infty$ off-orbit), $\hat{\omega}_W^L$ the BW vacuum as a character of $\mathcal{M}^L$, and $\hat{\mathcal{U}}^L$ the truncation cover. The Bridge Theorem [60, Theorem 6.4] verifies four properties (B1)–(B4) at the K$^+$-weak-form level.

The reverse triangle (B2) is the substantive ingredient. Its proof proceeds by Decomposition O on causal triples $(\hat{\omega}_x, \hat{\omega}_y, \hat{\omega}_z)$:

- *Case (i): all three on the same modular orbit.* On-orbit additivity of modular flow gives equality $\ell^L(\hat\omega_x, \hat\omega_y) + \ell^L(\hat\omega_y, \hat\omega_z) = \ell^L(\hat\omega_x, \hat\omega_z)$.
- *Cases (ii), (iv): at least one pair has $\ell^L = -\infty$.* The reverse triangle holds trivially via the Mondino–Sämann convention $\pm\infty \mp \infty = 0$.
- *Case (iii): three different orbits, all pairs causally related.* **Structurally empty at K$^+$-weak-form** because modular flow on the M-diagonal topography $\mathcal{M}^L$ preserves the (N, L)-labels: $[K^W_\alpha, M^{\text{spat}}_{N,L,0}] = 0$ since K^W_α has eigenvalue 0 on the $m_j = 0$ subspace. Hence any two characters related by modular flow share the same (N, L)-label, and three distinct-orbit characters cannot all be pairwise causally related.

The K$^+$-weak-form bridge closes (B2) via the structural emptiness of Case (iii). Paper 49 extends to the strict-strong-form by enlarging the topography to activate Case (iii) non-trivially.

2.2 The Paper 46 Appendix B enlarged substrate

Paper 46 [58] Appendix B introduces the chirality-flipping generators $M^{\text{flip}}_{N,L,M}$ in the chirality-asymmetric diagonal form

$$M^{\text{flip}}_{N,L,M} \;=\; \operatorname{diag}\!\big(W^{NLM},\; -W^{NLM}\big) \quad \text{in the chiral basis}\,. \tag{1}$$

The anti-commutator $\{\gamma^0, M^{\text{flip}}_{N,L,M}\} = 0$ is bit-exact zero in the chiral basis. The Avery–Wen–Avery (AWA) Weyl-spinor multiplier W^{NLM} [56] acts on the spinor bundle by multiplication by the spatial 3-Y function $Y^{(3)}_{N,L,M}$.

The *full enlarged topography* is

$$\mathcal{M}^{L,\text{flip}}_{\text{full}} \;:=\; \operatorname{span}_{\mathbb{C}}\!\Big(\mathcal{O}^L \cup \big\{M^{\text{flip}}_{N,L,M} \otimes I_{N_t} \;:\; N \le n_{\max},\; L < N,\; |M| \le L\big\}\Big) \subseteq \mathcal{A}^K, \tag{2}$$

where $\mathcal{O}^L$ is the Paper 44 natural Lorentzian operator system.

The $M = 0$ *chirality-flip topography* (the Phase-1 stepping stone) is

$$\mathcal{M}^{L,\text{flip}}_{M=0} \;:=\; \operatorname{span}_{\mathbb{C}}\!\Big(\mathcal{M}^L \cup \big\{M^{\text{flip}}_{N,L,0} \otimes I_{N_t} \;:\; N \le n_{\max},\; L < N\big\}\Big) \subseteq \mathcal{A}^K. \tag{3}$$

2.3 The Q1′-Light bimodal-Abelianness finding

The Q1′-Light diagnostic (Track 0[1] of the Q1′ arc) showed that the enlarged topography is structurally bimodal in M:

Case A ($M = 0$ enlargement): For W^{NL0} pure multiplication operators (no m_j-rotation content at $M = 0$ by real-spherical-harmonic parity), any two such operators commute pointwise, and the chirality-flip commutator reduces to

$$[M^{\text{flip}}_{N_1,L_1,0}, M^{\text{flip}}_{N_2,L_2,0}] = \operatorname{diag}\!\big([W^{N_1L_10}, W^{N_2L_20}],\, [W^{N_1L_10}, W^{N_2L_20}]\big) = 0$$

bit-exact. **Abelianness preserved.** The Gelfand spectrum of $\mathcal{M}^{L,\text{flip}}_{M=0}$ exists and decomposes as a two-fold $\mathbb{Z}/2$-cover $\widehat{\mathcal{M}^{L,\text{flip}}_{M=0}} = \widehat{\mathcal{M}^L}^{(+)} \sqcup \widehat{\mathcal{M}^L}^{(-)}$ (Phase-1 Theorem; recapped here as Theorem 4.2).

Case B (full $M \neq 0$ enlargement): For W^{NLM} with $M \neq 0$, the operator carries non-trivial Wigner-rotation content on the m_j labels, and $[W^{N_1L_1M_1}, W^{N_2L_2M_2}] \neq 0$ in general (Clebsch–Gordan recoupling). Hence $[M^{\text{flip}}_{N_1,L_1,M_1}, M^{\text{flip}}_{N_2,L_2,M_2}] \neq 0$ in general. **Abelianness breaks.** The Gelfand spectrum does not apply directly.

[1] Sprint/track designations are anchors into the project's internal chronicle (`CHANGELOG.md` in the repository).

The Phase-1 Case A enlargement compresses cleanly but does NOT activate Decomposition O Case (iii) (the chirality grading does not introduce new (N, L)-labels under modular flow). The substantive Q1′ content requires Case B (full $M \neq 0$), which forces the operator-system extension of the synthetic side. We pursue this via Option γ of the Q1′-Light diagnostic: replace the target category with OSLPLS, an operator-system Lorentzian pre-length space category that contains Mondino–Sämann as a faithful sub-category.

2.4 Mondino–Sämann covered Lorentzian pre-length spaces

We recall the Mondino–Sämann pre-length space axioms from [34].

Definition 2.1 (MS LPLS, Def 2.3 of [34])**.** A *Lorentzian pre-length space* (LPLS) is a triple (X, ℓ) where

(a) X is a set (underlying point set),

(b) X carries a topology finer than the chronological topology generated by causal-future sets,

(c) $\ell : X \times X \to \{-\infty\} \cup [0, \infty]$ is a time-separation function with $\ell(x, x) = 0$ for all $x \in X$,

(d) ℓ satisfies the reverse triangle inequality $\ell(x, y) + \ell(y, z) \leq \ell(x, z)$ on causal triples.

Definition 2.2 (Covered LPLS, Def 3.8 of [34])**.** A *covered Lorentzian pre-length space* is a 4-tuple $(X, \ell, o, \mathcal{U})$ where (X, ℓ) is an LPLS, $o \in X$ is the basepoint, and $\mathcal{U} = (U_k)_{k \in \mathbb{N}}$ is a countable cover such that $\bigcup_k U_k = X$, $U_k \subseteq U_{k+1}$, $o \in U_k$ for all k, and $\sup_{x,y \in U_k} \tau(x, y) < \infty$ (slab bound), with $\tau := \max(0, \ell)$.

The category $\mathbf{LorPLG}_{\mathrm{cov}}$ has objects covered LPLS, with morphisms ℓ-preserving basepoint-preserving cover-preserving maps ([34] Def 4.4).

2.5 The Connes–Rovelli thermal-time hypothesis and the Connes 1973 cocycle Radon–Nikodym derivative

Connes–Rovelli 1994 [11]. Let $\mathcal{M}$ be a von Neumann algebra. Let ω be a faithful normal state on $\mathcal{M}$. The Tomita–Takesaki theorem associates to ω a one-parameter modular automorphism group $\sigma_t^\omega : \mathcal{M} \to \mathcal{M}$ ($t \in \mathbb{R}$). The *thermal-time hypothesis* states that at equilibrium in ω at inverse temperature β, the modular flow σ_t^ω is the physical time evolution (up to the rescaling $t_{\mathrm{phys}} = t/\beta$).

For a single KMS state, the thermal time is canonically associated with ω via the GNS construction + Tomita–Takesaki modular operator. Different KMS states have different thermal times.

The Connes 1973 cocycle Radon–Nikodym derivative [9]. For any two faithful KMS states $\omega^{\sigma_1}, \omega^{\sigma_2}$ at the same inverse temperature β on the same von Neumann algebra $\mathcal{N}$, there exists a unitary cocycle $u_t^{\sigma_1\sigma_2} \in \mathcal{N}$ ($t \in \mathbb{R}$) called the Connes cocycle Radon–Nikodym derivative $[D\omega^{\sigma_2} : D\omega^{\sigma_1}]_t$, satisfying:

(C1) *Unitary:* $u_t^{\sigma_1\sigma_2}(u_t^{\sigma_1\sigma_2})^* = 1$.

(C2) *Cocycle identity (one-parameter):* $u_{t_1+t_2}^{\sigma_1\sigma_2} = u_{t_1}^{\sigma_1\sigma_2}\sigma_{t_1}^{\omega^{\sigma_1}}(u_{t_2}^{\sigma_1\sigma_2})$.

(C3) *Intertwining:* $\sigma_t^{\omega^{\sigma_2}}(a) = u_t^{\sigma_1\sigma_2}\sigma_t^{\omega^{\sigma_1}}(a)(u_t^{\sigma_1\sigma_2})^*$ for all $a \in \mathcal{N}$.

(C4) *Cocycle composition (chain rule):* $u_t^{\sigma_1\sigma_3} = u_t^{\sigma_1\sigma_2}\sigma_t^{\omega^{\sigma_1}}(u_t^{\sigma_2\sigma_3})$.

The cocycle is uniquely determined up to a "unitary-of-the-fixed-point-algebra" gauge that does not affect the modular flow intertwining at the automorphism level ([9] Cor 3.2).

The Bratteli–Robinson account [5], Theorem 5.3.10, gives the KMS-state structure under modular automorphism groups; the cocycle Radon–Nikodym derivative is the operator-algebraic measure-theoretic dual of the absolute-continuity of one KMS state with respect to another.

The gap. Connes–Rovelli 1994 addresses a *single* KMS state. The cross-KMS-state structure (the cocycle machinery, with its triple-intersection composition rule) is available but is not explicitly identified in the literature as a "thermal-time stack" suitable for the OSLPLS reverse triangle proof. Section 6 closes this gap by isolating the triple-intersection cocycle identity (TICI) as the stack-consistency property and applying it via Datta's max-divergence chain inequality to give the strict super-additivity.

3 The OSLPLS category

3.1 Axiom transport from MS to operator systems

We walk through each Mondino–Sämann axiom (§2.4) and identify its operator-system generalization. The classification is (i) transports directly (no commutativity used), (ii) transports with modification (commutativity used in a replaceable step), (iii) does not transport (commutativity is structurally load-bearing).

Theorem 3.1 (MS Axiom Transport)**.** *Let $\mathcal{M} \subseteq \mathcal{A}$ be an operator system (not necessarily Abelian) in a unital C^*-algebra $\mathcal{A}$, equipped with a faithful basepoint state $\omega \in \mathcal{S}(\mathcal{M})$ whose Tomita–Takesaki modular flow σ_t^ω on $\mathcal{M}$ is well-defined. Then:*

(a) Underlying set *(MS Def 2.3(a)): $X^{\mathrm{OS}} := \mathcal{S}(\mathcal{M})$ replaces MS's X. When $\mathcal{M}$ is Abelian, the pure-state subset $\widehat{\mathcal{M}}$ recovers the Gelfand spectrum; otherwise $\mathcal{S}(\mathcal{M})$ is the operator-system-natural generalization.*

(b) Topology *(MS Def 2.3(b)): weak-* on $\mathcal{S}(\mathcal{M})$ refines the chronological topology generated by operator-system causal-future sets $I^+_{\mathrm{OS}}(\omega) := \{\omega' \in \mathcal{S}(\mathcal{M}) : \ell^{\mathrm{OS}}(\omega,\omega') > 0\}$. Transports directly.*

(c) Time separation *(MS Def 2.3(c)): $\ell^{\mathrm{OS}}_{\omega_0} : \mathcal{S}(\mathcal{M}) \times \mathcal{S}(\mathcal{M}) \to \{-\infty\} \cup [0,\infty]$ via modular flow on the basepoint state:*

$$\ell^{\mathrm{OS}}_{\omega_0}(\omega',\omega'') := \begin{cases} \kappa_g \cdot \tau^{\omega_0}_{\mathrm{mod}}(\omega',\omega'') & \textit{if } \omega'' = \omega' \circ \sigma_t^{\omega_0} \textit{ for some } t \geq 0 \\ -\infty & \textit{otherwise}, \end{cases} \tag{4}$$

with $\tau^{\omega_0}_{\mathrm{mod}}(\omega',\omega'') := \inf\{t \geq 0 : \omega'' = \omega' \circ \sigma_t^{\omega_0}\}$ the modular precedence time and κ_g the BW surface gravity (canonical $\kappa_g = 1$, $\beta = 2\pi$).

(d) Reverse triangle inequality *(MS Def 2.3(d)): $\ell^{\mathrm{OS}}(\omega_x,\omega_y) + \ell^{\mathrm{OS}}(\omega_y,\omega_z) \leq \ell^{\mathrm{OS}}(\omega_x,\omega_z)$. The statement transports verbatim; the proof on the Krein-side bridge image is the substantive content of §6.*

(e) ε-net *(MS Def 3.2): operator-system causal diamonds $J^{\mathrm{OS}}(\omega_x,\omega_y) := \{\omega_z \in \mathcal{S}(\mathcal{M}) : \omega_x \preceq_{\mathrm{mod}} \omega_z \preceq_{\mathrm{mod}} \omega_y\}$ replace MS causal diamonds. Transports directly.*

(f) Covered LPLS *(MS Def 3.8): cover by operator-sub-systems $\mathcal{M}_k \subseteq \mathcal{M}$ with $U_k^{\mathrm{OS}} = \mathcal{S}(\mathcal{M}_k)$. Transports directly.*

(g) Morphisms *(MS Def 4.4): unital completely positive (UCP) maps $\phi : \mathcal{M}_2 \to \mathcal{M}_1$ with state-pullback basepoint preservation. Transports with modification: the morphism class shifts from set maps to UCP maps in the Connes–van Suijlekom convention [12].*

(h) LGH / pLGH convergence *(MS Defs 3.6, 3.12): per-cover operator-system LGH at each k. Transports directly.*

All eight MS axioms transport to the operator-system setting with the modifications enumerated above. None forces commutativity of $\mathcal{M}$.

Proof sketch. For (a): the state space $\mathcal{S}(\mathcal{M})$ of an operator system is weak-* compact when $\mathcal{M}$ unital. For Abelian $\mathcal{M}$, the pure states recover the Gelfand spectrum.

For (b): weak-* convergence implies pointwise evaluation on every multiplier, hence refines topologies generated by sub-collections of multipliers including I^+_{OS}.

For (c): $\sigma_t^{\omega_0}$ is functorial in $(\mathcal{M}, \omega_0)$ via Tomita–Takesaki, hence well-defined on any operator system with faithful basepoint state.

For (d): the inequality on $\{-\infty\} \cup [0, \infty]$ is formally commutativity-blind. The MS proof uses the twin-paradox commutative geometric argument; the operator-system proof requires the cocycle-stack construction of §6.

For (e), (f), (h): set-theoretic operations on $\mathcal{S}(\mathcal{M})$ and $(\mathcal{M}_k)$ once ℓ^{OS} is fixed.

For (g): UCP composition is standard for operator systems [12]; basepoint-state pullback is well-defined for any UCP map; modular intertwining is automatic from state pullback (Lemma 3.5).

Substantive observation: axiom (d) is the only MS axiom whose *proof* uses commutativity structurally. All other modifications are operator-system-natural Connes–vS framework substitutions. □

3.2 The OSLPLS object

Definition 3.2 (Operator-System Lorentzian Pre-Length Space)**.** An *operator-system Lorentzian pre-length space* (OSLPLS) is a 5-tuple

$$\mathfrak{X}^{\mathrm{OS}} = (\mathcal{A}, L, \mathcal{M}, \omega, \mathcal{U})$$

where:

(A1) $\mathcal{A}$ is a separable unital C*-algebra (the underlying algebra).

(A2) $L : \mathrm{dom}(L) \subseteq \mathcal{A} \to [0, \infty]$ is a lower semicontinuous Lipschitz seminorm with $\mathrm{dom}(L)$ dense in $\mathcal{A}$ and $L(a) = 0 \iff a \in \mathbb{C} \cdot 1$. Equivalently, L generates a Latrémolière PPQMS structure with K$^+$-positivity boundedness [25, 60].

(A3) $\mathcal{M} \subseteq \mathcal{A}$ is an **operator system** (not necessarily Abelian), unital, *-closed, with $L|_{\mathcal{M}}$ inheriting the Lipschitz structure.

(A4) $\omega \in \mathcal{S}(\mathcal{M})$ is a faithful (GNS-cyclic) state, generating a well-defined Tomita–Takesaki modular flow σ_t^ω on $\mathcal{M}$.

(A5) $\mathcal{U} = (\mathcal{M}_k)_{k\in\mathbb{N}}$ is a sequence of operator-sub-systems $\mathcal{M}_k \subseteq \mathcal{M}$ satisfying $\bigcup_k \mathcal{M}_k = \mathcal{M}$ (dense), $\mathcal{M}_k \subseteq \mathcal{M}_{k+1}$ (nested), $\omega|_{\mathcal{M}_k}$ faithful for all k, and $\sup_{\omega',\omega''\in\mathcal{S}(\mathcal{M}_k)} \tau^{\mathrm{OS}}(\omega', \omega'') \leq \beta_k < \infty$ (slab bound), with $\tau^{\mathrm{OS}} := \max(0, \ell^{\mathrm{OS}}_\omega)$.

Remark 3.3. The OSLPLS object class is *structurally constrained, not freely chosen.* The three design choices — state-space-as-underlying-set ($X \to \mathcal{S}(\mathcal{M})$), modular-flow-as-time-separation ($\ell \to \ell^{\mathrm{OS}}$), UCP morphisms — are all forced by Theorem 3.1: requiring that (a) OSLPLS contain commutative MS as a faithful sub-category and (b) admit the Krein-side bridge image leaves essentially no design freedom.

3.3 The OSLPLS morphism

Definition 3.4 (OSLPLS morphism)**.** Let $\mathfrak{X}_1^{\mathrm{OS}}, \mathfrak{X}_2^{\mathrm{OS}}$ be two OSLPLS objects. An *OSLPLS morphism* $\phi : \mathfrak{X}_1^{\mathrm{OS}} \to \mathfrak{X}_2^{\mathrm{OS}}$ is a unital completely positive (UCP) map $\phi : \mathcal{A}_2 \to \mathcal{A}_1$ (contravariant) such that:

(M1) $\phi(\mathcal{M}_2) \subseteq \mathcal{M}_1$ (topography preservation),

(M2) $L_1(\phi(a)) \leq L_2(a)$ (Lipschitz contractivity),

(M3) $\omega_1 \circ \phi = \omega_2$ (basepoint pullback),

(M4) $\phi(\mathcal{M}_{2,k}) \subseteq \mathcal{M}_{1,k}$ for all k (cover preservation),

(M5) $\phi \circ \sigma_t^{\omega_2} = \sigma_t^{\omega_1} \circ \phi$ on $\mathcal{M}_2$ for all t (modular intertwining).

Lemma 3.5 (Modular intertwining automaticity)**.** *If $\phi : \mathcal{A}_2 \to \mathcal{A}_1$ is a UCP map satisfying (M3) and ω_2 is faithful on $\mathcal{M}_2$, then (M5) is automatic.*

Proof. The Tomita–Takesaki modular flow is functorial in the (operator system, basepoint state) pair: a UCP map with state pullback $\phi^*(\omega_1) = \omega_2$ induces (via Stinespring dilation) a unitary $V_\phi : \mathcal{H}_{\omega_2} \to \mathcal{H}_{\omega_1}$ between GNS spaces, intertwining the modular operators $V_\phi \Delta_{\omega_2}^{it} = \Delta_{\omega_1}^{it} V_\phi$. The algebra-level intertwining follows. □

Definition 3.6 (The category $\mathbf{OSLPLS}_{\mathrm{cov}}$)**.** The category $\mathbf{OSLPLS}_{\mathrm{cov}}$ has objects OSLPLS 5-tuples (Definition 3.2), morphisms UCP maps with (M1)–(M5) (Definition 3.4), and contravariant UCP composition. Identity morphisms exist; composition is associative; all axioms preserved under composition.

3.4 Commutative MS as a faithful sub-category

Theorem 3.7 (Embedding functor ι)**.** *Define $\iota : \mathbf{LorPLG}_{\mathrm{cov}} \to \mathbf{OSLPLS}_{\mathrm{cov}}$ by*

$$\iota(X, \ell, o, \mathcal{U}) := (C(X), L_\ell, C(X), \delta_o, (C(\overline{U_k}))_{k\in\mathbb{N}})$$

on objects, and $\iota(f) := f^* : C(X_2) \to C(X_1)$, $\iota(f)(g) := g \circ f$ on morphisms. Then ι is a well-defined functor satisfying:

(a) Object correspondence: $\iota(X, \dots)$ verifies (A1)–(A5).

(b) Morphism correspondence: $\iota(f)$ verifies (M1)–(M5).

(c) Faithful: distinct MS objects map to non-isomorphic OSLPLS objects.

(d) Full on image: any OSLPLS morphism between objects in $\iota(\mathbf{LorPLG}_{\mathrm{cov}})$ comes from a unique MS isometry.

Proof sketch. (a) $C(X)$ is separable for X metrizable compact Hausdorff; L_ℓ is lower semicontinuous with kernel $\mathbb{C} \cdot 1$ and dense Lipschitz subset; $C(X)$ is its own (trivially Abelian) operator system; δ_o is faithful in the GNS-cyclic sense on the operator system generated by functions near o; the nested cover and slab bound inherit from MS.

(b) f^* is UCP (any $*$-homomorphism between Abelian C*-algebras is UCP); topography preservation automatic; Lipschitz contractivity $L_{\ell_1}(f^* g) \leq L_{\ell_2}(g)$ uses ℓ-preservation; basepoint pullback $(\delta_{o_1} \circ f^*)(g) = g(o_2) = \delta_{o_2}(g)$; cover preservation from MS; modular intertwining trivial because modular flow on Abelian C*-algebras is the identity.

(c) An OSLPLS isomorphism between ι images corresponds (via Gelfand–Naimark) to a homeomorphism of spectra preserving Lipschitz seminorm, basepoint Dirac state, and cover; hence to an MS isometry.

(d) Any OSLPLS morphism in the image of ι corresponds (via Gelfand–Naimark) to a continuous map f preserving ℓ, basepoint, and cover; hence $\phi = \iota(f)$. □

Remark 3.8. On the commutative MS sub-category, the modular flow $\sigma_t^{\delta_o}$ is trivial on Abelian $C(X)$, so the operator-system time separation $\ell_{\delta_o}^{\mathrm{OS}}$ defined by (4) is degenerate. The MS time separation ℓ is *not* recovered as $\ell_{\delta_o}^{\mathrm{OS}}$; it is stored separately as the Lipschitz seminorm L_ℓ, with the duality

$$\ell(x,y) = \sup_{f\in C(X),\, L_\ell(f)\leq 1} |f(x)-f(y)|\cdot \mathrm{sign}(\text{causal direction})$$

recovering ℓ from L_ℓ via the standard metric/Lipschitz duality.

The substantive content of OSLPLS — non-trivial modular flow generating non-trivial causal structure on $\mathcal{S}(\mathcal{M})$ — is exclusively a non-commutative feature. On the commutative sub-category, modular flow is trivial and causal structure must be supplied externally via L_ℓ; on non-commutative $\mathcal{M}$, causal structure emerges from the algebraic data.

The connection to the Connes–Rovelli thermal-time hypothesis (§2.5): when the topography is non-commutative, thermal time generates geometric causal structure intrinsically. The Wick rotation identifies these as Lorentzian causal structure in the limit.

4 The Krein-side bridge candidate W^{flip}

4.1 Phase-1 chirality-graded stepping stone

We first recall the $M=0$ enlarged construction (Phase-1 of the Q1′ arc) as a structural stepping stone, then extend to the full $M\neq 0$ enlargement in §4.2.

Lemma 4.1 (Krein $M=0$-chirality-flip topography)**.** *The $*$-subalgebra $\mathcal{M}^{L,\mathrm{flip}}_{M=0}$ of $\mathcal{A}^K$ defined by (3) is a Krein topography of $(\mathcal{A}^K, L^K)$. Specifically:*

(a) $\mathcal{M}^{L,\mathrm{flip}}_{M=0}$ is Abelian (Q1′-Light §2.3).

(b) It contains the strictly positive element $h^L = K^W_\alpha/Z \in \mathcal{M}^L \subseteq \mathcal{M}^{L,\mathrm{flip}}_{M=0}$.

(c) The BW vacuum ω^L_W restricts to a character witnessing the boundedness axiom.

(d) The truncated projector sequence $h_n = P_{(n_{\max}(n),N_t(n),T(n))}$ is a Krein-Lipschitz ω^L_W-pinned exhaustive sequence in $\mathcal{M}^{L,\mathrm{flip}}_{M=0}$.

Moreover, $\mathcal{M}^{L,\mathrm{flip}}_{M=0}$ carries a $\mathbb{Z}/2$-grading with the chirality-flip generators in the odd-degree subspace and the M-diagonal scalar generators in the even-degree subspace.

Theorem 4.2 (Two-fold-cover Gelfand spectrum)**.** *The Gelfand spectrum of $\mathcal{M}^{L,\mathrm{flip}}_{M=0}$ decomposes as a free two-fold $\mathbb{Z}/2$-cover:*

$$\widehat{\mathcal{M}^{L,\mathrm{flip}}_{M=0}} = \widehat{\mathcal{M}^L}^{(+)} \sqcup \widehat{\mathcal{M}^L}^{(-)}, \tag{5}$$

*with each sheet $\widehat{\mathcal{M}^L}^{(\pm)}$ in bijection with $\widehat{\mathcal{M}^L}$ via restriction to the M-diagonal even-graded subalgebra. The chirality-grading sign $\epsilon\in\{+1,-1\}$ specifies $\chi^{(\epsilon)}(M^{\mathrm{flip}}_{N,L,0}) = \epsilon\cdot\chi(M^{\mathrm{spat}}_{N,L,0})$. The BW vacuum ω^L_W restricts to the $\mathbb{Z}/2$-**invariant** basepoint (the average of the two pure characters above the K^+-weak-form character) because $\omega^L_W(M^{\mathrm{flip}}_{N,L,0}) = 0$ by chirality-symmetric thermal density at $m_j = 0$.*

Proof sketch. Multiplicativity of characters forces $\chi(M^{\mathrm{flip}}_{N,L,0})^2 = \chi(M^{\mathrm{spat}}_{N,L,0})^2$, hence $\chi(M^{\mathrm{flip}}_{N,L,0}) = \pm\chi(M^{\mathrm{spat}}_{N,L,0})$. The cross-product rule (the diagonal anti-symmetric signs cancel in $M^{\mathrm{flip}}_{N_1,L_1,0}\cdot M^{\mathrm{flip}}_{N_2,L_2,0} = M^{\mathrm{spat}}_{N_1,L_1,0}\cdot M^{\mathrm{spat}}_{N_2,L_2,0}$) forces a global sign choice $\epsilon_1\cdot\epsilon_2 = +1$ on the chirality-flip generators. Hence each $\chi^L\in\widehat{\mathcal{M}^L}$ lifts to exactly two characters $\chi^{(+)},\chi^{(-)}$. The BW vacuum sits at the $\mathbb{Z}/2$-fixed point because the wedge KMS density at $m_j=0$ is chirality-symmetric. □

The Phase-1 chirality-graded bridge functor $W^{\mathrm{flip},M=0}$: $\mathbf{KreinMetaMet}_{\mathrm{pp}}^{\mathrm{flip},M=0} \to \mathbf{LorPLG}_{\mathrm{cov}}^{\mathbb{Z}/2}$ extends the A.3′ Wick-rotation functor with a $\mathbb{Z}/2$-cover structure at no substantial cost. **Phase-1 explicitly does not activate Decomposition O Case (iii)**: at $M = 0$, $[K_\alpha^W, M_{N,L,0}^{\mathrm{flip}}] = 0$ trivially (since K_α^W vanishes on the $m_j = 0$ subspace), so modular flow preserves the chirality grading, and three different modular orbits cannot have all three pairs causally related.

Phase-1 is therefore a structurally-additive stepping stone establishing machinery (chirality-graded bridge functor, two-fold-cover Gelfand spectrum, $\mathbb{Z}/2$-graded LPLS target category) but *not* substantively new content. The substantive content of the Q1′ question requires the full $M \neq 0$ enlargement, which is the subject of §4.2 below.

4.2 W^{flip} on the full $M \neq 0$ enlargement: objects

The full enlarged Krein-PPQMS source object is

$$\mathbb{X}^{K,\mathrm{flip},\mathrm{full}} := (\mathcal{A}^K, L^K, \mathcal{M}_{\mathrm{full}}^{L,\mathrm{flip}}, \omega_W^L), \tag{6}$$

with $\mathcal{M}_{\mathrm{full}}^{L,\mathrm{flip}}$ from (2) the full chirality-flip enlarged topography (non-Abelian at $M \neq 0$ per Q1′-Light §2.3).

Definition 4.3 (Bridge functor W^{flip} on objects)**.** Define W^{flip} : $\mathbf{KreinMetaMet}_{\mathrm{pp}}^{\mathrm{flip},\mathrm{full}} \to \mathbf{OSLPLS}_{\mathrm{cov}}$ on objects by

$$W^{\mathrm{flip}}(\mathbb{X}^{K,\mathrm{flip},\mathrm{full}}) := (\mathcal{A}^K, L^K, \mathcal{M}_{\mathrm{full}}^{L,\mathrm{flip}}, \omega_W^L|_{\mathcal{M}_{\mathrm{full}}^{L,\mathrm{flip}}}, \mathcal{U}_{\mathrm{OS}}^{L,\mathrm{flip}}) \tag{7}$$

where:

- The ambient C*-algebra, Lipschitz seminorm, and topography are inherited from the source (no Gelfand-spectrum step).
- The basepoint state is the restriction $\omega := \omega_W^L|_{\mathcal{M}_{\mathrm{full}}^{L,\mathrm{flip}}}$.
- The cover is the operator-system truncated cover $\mathcal{U}_{\mathrm{OS}}^{L,\mathrm{flip}} = (\mathcal{M}_{\mathrm{full},k}^{L,\mathrm{flip}})_{k\in\mathbb{N}}$ where $\mathcal{M}_{\mathrm{full},k}^{L,\mathrm{flip}}$ is the restriction of the full enlarged topography to the k-th truncation $(n_{\max}(k), N_t(k), T(k))$.

Note. This is not a Gelfand-spectrum construction; the source topography is non-Abelian, so the OSLPLS object directly stores the operator-system structure with the state space $\mathcal{S}(\mathcal{M}_{\mathrm{full}}^{L,\mathrm{flip}})$ playing the role of underlying set.

Theorem 4.4 (Krein-side bridge candidate verifies OSLPLS axioms)**.** *The 5-tuple $W^{\mathrm{flip}}(\mathbb{X}^{K,\mathrm{flip},\mathrm{full}})$ defined in* (7) *satisfies all five OSLPLS object axioms (A1)–(A5) of Definition 3.2.*

Proof. **(A1) Ambient C*-algebra.** $\mathcal{A}^K$ is separable (finite-dimensional at finite cutoff per Paper 44 [56]; norm-closure of finite-dimensional in the continuum limit per Paper 47 [59]). ✓

(A2) Lipschitz seminorm. $L^K(a) = \|[D_L, a]\|_{\mathrm{op}}$ is lower semicontinuous; $\mathrm{dom}(L^K)$ dense in $\mathcal{A}^K$; $L^K(a) = 0$ iff $a \in \mathbb{C}\cdot 1$ (since D_L has cyclic vector on $\mathcal{K}^+$ per Paper 42 [54]); K$^+$-positivity boundedness holds per Paper 46 [58]. ✓

(A3) Topography. $\mathcal{M}_{\mathrm{full}}^{L,\mathrm{flip}}$ is an operator system in $\mathcal{A}^K$: contains the unit; $*$-closed (via $(M_{N,L,M}^{\mathrm{flip}})^* = M_{N,L,-M}^{\mathrm{flip}}$ from $\overline{Y_{N,L,M}^{(3)}} = (-1)^M Y_{N,L,-M}^{(3)}$); closed under linear combinations by construction; non-Abelian at $M \neq 0$ (Q1′-Light §2.3 Case B); $L^K|_{\mathcal{M}_{\mathrm{full}}^{L,\mathrm{flip}}}$ inherits Lipschitz structure. ✓

(A4) Basepoint state. $\omega = \omega_W^L|_{\mathcal{M}_{\mathrm{full}}^{L,\mathrm{flip}}}$ is faithful (GNS-cyclic) because ω_W^L is faithful on $\mathcal{A}^K$ at finite cutoff (the wedge BW vacuum is the cyclic-separating vector for the wedge von Neumann algebra per Paper 42 [54] §5). At $M \neq 0$, $\omega_W^L(M_{N,L,M}^{\mathrm{flip}}) \neq 0$ in general (no symmetry forces vanishing), so the chirality-flip subspace is not in the kernel and GNS-cyclic faithfulness holds. The modular flow $\sigma_t^{\omega_W^L} = \mathrm{Ad}(e^{itK_\alpha^W})$ restricts to the enlarged topography, with non-trivial action on chirality-flip generators at $M \neq 0$ because $[K_\alpha^W, M_{N,L,M}^{\text{flip}}] \neq 0$ when the BW polar reflection mixes m_j labels. ✓

(A5) Operator-system cover. $\mathcal{U}_{\text{OS}}^{L,\text{flip}} = (\mathcal{M}_{\text{full},k}^{L,\text{flip}})_{k\in\mathbb{N}}$: dense union via admissible-scaling cutoff $(n_{\max}(k), N_t(k), T(k)) \to (\infty, \infty, T_\infty)$ [59]; nested via cutoff increase; BW vacuum extends to each truncation (M-block-diagonal projector + symmetry); slab bound $\sup \tau^{\text{OS}} \leq \beta_k = 2\pi$ via Paper 42 [54] §5 integer spectrum of K_α^W (modular orbits are 2π-periodic in canonical $\kappa_g = 1$). ✓ □

4.3 W^{flip} on morphisms

For source morphisms (topographic Krein M-isometries on the enlarged substrate), W^{flip} acts via contravariant UCP pullback in the Connes–vS convention [12].

Definition 4.5 (Topographic Krein M-isometry on enlarged substrate)**.** Let $\mathbb{X}_1^{K,\text{flip,full}}, \mathbb{X}_2^{K,\text{flip,full}}$ be two enlarged-substrate Krein PPQMS objects. A *topographic Krein M-isometry on the enlarged substrate* $\pi^{K,\text{flip}} : \mathbb{X}_1^{K,\text{flip,full}} \to \mathbb{X}_2^{K,\text{flip,full}}$ is a proper $*$-epimorphism $\pi^{K,\text{flip}} : \mathcal{A}_1^K \twoheadrightarrow \mathcal{A}_2^K$ satisfying:

(S1) $L_2^K(\pi^{K,\text{flip}}(a)) \leq L_1^K(a)$ (Lipschitz compatibility),

(S2) $\pi^{K,\text{flip}}(\mathcal{M}_{\text{full},1}^{L,\text{flip}}) \subseteq \mathcal{M}_{\text{full},2}^{L,\text{flip}}$ (enlarged topography preservation),

(S3) $\omega_{W,2}^L \circ \pi^{K,\text{flip}} = \omega_{W,1}^L$ (pin state preservation),

(S4) $\pi^{K,\text{flip}}(\mathcal{M}_{\text{full},1,k}^{L,\text{flip}}) \subseteq \mathcal{M}_{\text{full},2,k}^{L,\text{flip}}$ for all k (cover preservation).

Definition 4.6 (Bridge functor W^{flip} on morphisms)**.** For $\pi^{K,\text{flip}} : \mathbb{X}_1^{K,\text{flip,full}} \to \mathbb{X}_2^{K,\text{flip,full}}$, set

$$W^{\text{flip}}(\pi^{K,\text{flip}}) := (\pi^{K,\text{flip}})^* : W^{\text{flip}}(\mathbb{X}_2^{K,\text{flip,full}}) \to W^{\text{flip}}(\mathbb{X}_1^{K,\text{flip,full}}) \tag{8}$$

where $(\pi^{K,\text{flip}})^*$ is the algebraic action $\pi^{K,\text{flip}}$ viewed as a UCP map $\mathcal{A}_2^K \to \mathcal{A}_1^K$ in the OSLPLS contravariant direction (a category- theoretic direction-of-arrow flip, not an operation on the algebra map).

Theorem 4.7 (Functoriality of W^{flip})**.** $W^{\text{flip}} : \mathbf{KreinMetaMet}_{\text{pp}}^{\text{flip,full}} \to \mathbf{OSLPLS}_{\text{cov}}$ *defined by Definition 4.3 (objects) and Definition 4.6 (morphisms) is a well-defined functor. Specifically:*

(a) For every source morphism $\pi^{K,\text{flip}}$, $W^{\text{flip}}(\pi^{K,\text{flip}})$ is an OSLPLS morphism (verifies M1–M5 of Definition 3.4).

(b) Identity preservation: $W^{\text{flip}}(\mathrm{id}_{\mathbb{X}^{K,\text{flip,full}}}) = \mathrm{id}_{W^{\text{flip}}(\mathbb{X}^{K,\text{flip,full}})}$.

(c) Contravariant composition: $W^{\text{flip}}(\pi_2 \circ \pi_1) = W^{\text{flip}}(\pi_1) \circ W^{\text{flip}}(\pi_2)$.

Proof sketch. (a) M1–M4 follow from S1–S4 read in the contravariant direction (set-theoretic image inclusion in the covariant direction equals the preimage inclusion in the contravariant direction). M5 is automatic from M3 (Lemma 3.5).

(b) The identity $*$-homomorphism is self-dual.

(c) Standard contravariant $*$-homomorphism composition: $(\pi_2 \circ \pi_1)^* = \pi_1^* \circ \pi_2^*$, matching OSLPLS contravariant composition. □

Theorem 4.8 (Commutative-case restriction agreement)**.** *The morphism action of W^{flip} restricted to the commutative MS sub-category via the embedding ι of Theorem 3.7 agrees with the A.3′ Wick-rotation functor* $\mathbf{W}$ *of Paper 48 [60] up to natural isomorphism:*

$$W^{\text{flip}}|_{\iota(\mathbf{LorPLG}_{\text{cov}})} \cong \mathbf{W} \circ \iota^{-1}.$$

Proof sketch. For a commutative source morphism π^K on Abelian $\mathcal{M}^L$, the A.3′ action $\mathbf{W}(\pi^K) = \hat{\pi}^K : \hat{\mathcal{M}}_2^L \to \hat{\mathcal{M}}_1^L$ is the dual Gelfand spectrum map. Restricting W^{flip} via ι gives the UCP pullback $(\pi^K)^* : C(\hat{\mathcal{M}}_2^L) \to C(\hat{\mathcal{M}}_1^L)$, which by Gelfand–Naimark identifies with $\hat{\pi}^K$ contravariantly. Hence the restriction agrees with A.3′. □

5 Off-orbit structural activation at full $M \neq 0$

The substantive new content of the Q1′ strong-form bridge sits in the modular flow structure on the enlarged topography. We isolate what changes between Phase-1's $M = 0$ enlargement and Phase-2's full $M \neq 0$ enlargement.

5.1 Why Phase-1 ($M = 0$) fails to activate Decomposition O Case (iii)

At $M = 0$, the BW Hamiltonian $K_\alpha^W = J_{\text{polar}}^W$ has integer spectrum $2m_j$ on the spinor bundle and vanishes on the $m_j = 0$ subspace. Hence

$$[K_\alpha^W, M_{N,L,0}^{\text{flip}}] \;=\; \text{diag}\big([K_\alpha^W|_{\mathcal{H}^+}, W^{NL0}],\, -[K_\alpha^W|_{\mathcal{H}^-}, W^{NL0}]\big) \;=\; 0,$$

since the polar reflection on $m_j = 0$ states acts trivially. The modular flow $\sigma_t^{\omega_W^L}$ acts trivially on the $M = 0$ chirality-flip generators (in addition to acting trivially on the M-diagonal scalar generators per Paper 48 [60]). Hence modular flow preserves both the (N, L) labels AND the chirality grading sign ϵ:

$$\hat{\sigma}_t^{\omega_W^L}(\chi^{(\epsilon)}) = \chi^{(\epsilon)} \quad \forall t, \forall \chi^{(\epsilon)} \in \widehat{\mathcal{M}_{M=0}^{L,\text{flip}}}. \tag{9}$$

Three distinct-orbit characters $\chi_1^{(\epsilon_1)}, \chi_2^{(\epsilon_2)}, \chi_3^{(\epsilon_3)}$ require three different (N, L, ϵ)-tuples, which over-constrains the causal-pair structure — in particular, modular flow preserves orbit labels, so two characters on different orbits cannot be causally related via modular flow on the M-diagonal topography. Hence Decomposition O Case (iii) is structurally empty under the $M = 0$ chirality-flip enlargement.

5.2 Why Phase-2 (full $M \neq 0$) activates Case (iii)

At $M \neq 0$, the situation is qualitatively different. The chirality-flip generator $M_{N,L,M}^{\text{flip}} = \text{diag}(W^{NLM}, -W^{NLM})$ involves the spatial multiplier W^{NLM}, which at $M \neq 0$ is *not* a multiplication operator by a real function: the AWA Y-function $Y_{N,L,M}^{(3)}$ carries non-trivial m_j-rotation content (Wigner D-matrix structure on the spinor bundle).

The BW Hamiltonian K_α^W is the polar-reflection generator, which mixes the m_j labels (the polar reflection $m_j \mapsto -m_j$ is a Z_2 involution on the spinor bundle, with the J_{polar} generator of the BW boost orbit acting non-trivially on the chirality-flip generators at $M \neq 0$):

$$[K_\alpha^W, M_{N,L,M}^{\text{flip}}] \;\neq\; 0 \quad \text{for } M \neq 0\,. \tag{10}$$

The modular flow $\sigma_t^{\omega_W^L} = \text{Ad}(e^{itK_\alpha^W})$ has non-trivial action on the chirality-flip generators at $M \neq 0$: $\sigma_t^{\omega_W^L}(M_{N,L,M}^{\text{flip}}) \neq M_{N,L,M}^{\text{flip}}$ generically. Different states on the operator-system state space $\mathcal{S}(\mathcal{M}_{\text{full}}^{L,\text{flip}})$ can sit on *genuinely different* modular orbits with non-trivial ℓ^{OS} relations on each orbit-segment.

5.3 Three-orbit triples and the off-orbit case

Decomposition O Case (iii) becomes structurally non-empty: there exist triples $(\omega_x, \omega_y, \omega_z) \in \mathcal{S}(\mathcal{M}^{L,\text{flip}}_{\text{full}})^3$ on three distinct modular orbits $\mathcal{O}^{\sigma_1}, \mathcal{O}^{\sigma_2}, \mathcal{O}^{\sigma_3}$ with all three pairs causally related (in the operator-system modular precedence sense). For such triples, the OSLPLS reverse triangle inequality

$$\ell^{\text{OS}}(\omega_x, \omega_y) + \ell^{\text{OS}}(\omega_y, \omega_z) \;\le\; \ell^{\text{OS}}(\omega_x, \omega_z) \tag{11}$$

must hold with *strict* inequality (in the substantive case where the three orbits are pairwise distinct). The proof requires operator-algebraic machinery beyond the on-orbit modular flow additivity of Paper 42 [54] four-witness theorem; it is the substantive content of §6.

Remark 5.1. Physically, (11) on three-orbit triples is the operator-algebraic dual of the special-relativistic twin paradox: a worldline that detours through an intermediate KMS state ω^{σ_2} before reaching ω^{σ_3} accumulates *less* proper time than a direct worldline from ω^{σ_1} to ω^{σ_3}. The classical proof (e.g. Mondino–Sämann [34]) uses the geometric twin-paradox argument on commutative pre-length spaces. The operator-algebraic proof in §6 replaces the geometric argument with the information-theoretic content of the Connes–Rovelli thermal-time stack and Datta's max-divergence chain inequality.

6 The Connes–Rovelli thermal-time stack across distinct KMS states

This is the substantive new mathematics of the paper. We construct the Connes–Rovelli thermal-time stack across distinct KMS states in three steps and use it to prove strict super-additivity of ℓ^{OS}.

6.1 The orbit-KMS lemma

Lemma 6.1 (Orbit-KMS Lemma)**.** *Let $W^{\text{flip}}(\mathbb{X}^{K,\text{flip},\text{full}}) = (\mathcal{A}^K, L^K, \mathcal{M}^{L,\text{flip}}_{\text{full}}, \omega, \mathcal{U}^{L,\text{flip}}_{\text{OS}})$ be the Krein-side OSLPLS object of Theorem 4.4. For any modular orbit $\mathcal{O}^\sigma \subseteq \mathcal{S}(\mathcal{M}^{L,\text{flip}}_{\text{full}})$ (an orbit $\{\omega_0 \circ \sigma^\omega_t : t \in \mathbb{R}\}$ of some base state $\omega_0 \in \mathcal{S}(\mathcal{M}^{L,\text{flip}}_{\text{full}})$ under the modular flow σ^ω_t), there exists a canonical* effective KMS state *ω^σ on the orbit-generated sub-operator-system $\mathcal{M}^\sigma \subseteq \mathcal{M}^{L,\text{flip}}_{\text{full}}$ (the smallest $*$-closed σ^ω_t-invariant subspace of $\mathcal{M}^{L,\text{flip}}_{\text{full}}$ containing the orbit-generators) at inverse temperature $\beta = 2\pi$ such that:*

(a) $\omega^\sigma = \omega|_{\mathcal{M}^\sigma}$ is faithful on $\mathcal{M}^\sigma$;

(b) The Tomita–Takesaki modular flow $\sigma^{\omega^\sigma}_t$ on $\mathcal{M}^\sigma$ agrees with the restriction of σ^ω_t to $\mathcal{M}^\sigma$;

(c) ω^σ inherits the KMS condition from ω at the orbit level: for all $a, b \in \mathcal{M}^\sigma$,

$$\omega^\sigma\big(a\, \sigma^{\omega^\sigma}_{t+i\beta}(b)\big) = \omega^\sigma\big(\sigma^{\omega^\sigma}_t(b)\, a\big). \tag{12}$$

Proof. **(a) Faithfulness.** ω is faithful on $\mathcal{M}^{L,\text{flip}}_{\text{full}}$ (A4 of Theorem 4.4). The restriction $\omega|_{\mathcal{M}^\sigma}$ is faithful provided $\mathcal{M}^\sigma$ is non-trivial (which it is by construction, containing the orbit-generators). For $a \in \mathcal{M}^\sigma$ with $\omega^\sigma(a^*a) = 0$, we have $\omega(a^*a) = 0$, hence $a = 0$ by faithfulness of ω.

(b) Modular flow agreement. $\mathcal{M}^\sigma$ is σ^ω_t-invariant by construction. By the functoriality of the Tomita–Takesaki construction under sub-system restriction (a standard result in modular theory: if $\mathcal{N} \subseteq \mathcal{M}$ is a σ^ω_t-invariant sub-von-Neumann-algebra and $\omega|_{\mathcal{N}}$ is faithful, then $\sigma^{\omega|_{\mathcal{N}}}_t$ on $\mathcal{N}$ equals $\sigma^\omega_t|_{\mathcal{N}}$), the Tomita-Takesaki modular flow $\sigma^{\omega^\sigma}_t$ on $\mathcal{M}^\sigma$ equals $\sigma^\omega_t|_{\mathcal{M}^\sigma}$.

(c) KMS condition. The KMS condition for ω on $\mathcal{M}^{L,\text{flip}}_{\text{full}}$ at $\beta = 2\pi$ holds because ω is the BW vacuum restriction and Paper 42 [54] §5 establishes that the BW vacuum is a $(\sigma^\omega, \beta = 2\pi)$-KMS state on $\mathcal{A}^K$. Restricting both sides of the KMS identity to $\mathcal{M}^\sigma$ via (b) gives (12) for ω^σ. □

Structural reading. Lemma 6.1 establishes that each modular orbit on the OSLPLS bridge image carries an *intrinsic* effective KMS structure inherited from the global BW vacuum. The orbit-generated sub-operator-system $\mathcal{M}^\sigma$ behaves like a stand-alone KMS system at $\beta = 2\pi$. This unlocks the cross-KMS-state Connes 1973 cocycle machinery: each modular orbit becomes a labelled KMS system, and cross-orbit comparisons are cross-KMS-state comparisons.

6.2 The Connes cocycle Radon–Nikodym intertwiner

Per Lemma 6.1, each modular orbit $\mathcal{O}^\sigma$ has effective KMS state ω^σ on $\mathcal{M}^\sigma$. For two distinct orbits $\mathcal{O}^{\sigma_1}, \mathcal{O}^{\sigma_2}$ with overlap $\mathcal{M}^{\sigma_1} \cap \mathcal{M}^{\sigma_2} \neq \emptyset$ (generic for full $M \neq 0$ enlargement since the chirality-flip generators have shared spatial components across orbits), the Connes cocycle [9] $u_t^{\sigma_1\sigma_2} = [D\omega^{\sigma_2} : D\omega^{\sigma_1}]_t$ is well-defined on $\mathcal{M}^{\sigma_1} \cap \mathcal{M}^{\sigma_2}$ and intertwines the two modular flows.

The cross-orbit *thermal-time bridge* is

$$t^{\sigma_1\to\sigma_2}_{\text{therm}}(\omega'', \omega') := \tau^{\omega^{\sigma_2}}_{\text{mod}}(\omega'', \omega' \cdot u_t^{\sigma_1\sigma_2}), \tag{13}$$

where $\omega'' \in \mathcal{O}^{\sigma_2}$, $\omega' \in \mathcal{O}^{\sigma_1}$, and the product $\omega' \cdot u_t^{\sigma_1\sigma_2}$ uses the cocycle to map ω' from $\mathcal{O}^{\sigma_1}$-frame into $\mathcal{O}^{\sigma_2}$-frame. The intertwining property (C3) ensures that the cross-orbit thermal time is well-defined modulo the cocycle gauge (a unitary-of-fixed-point-algebra ambiguity that does not affect the modular flow's action on $\mathcal{S}(\mathcal{M})$).

6.3 The triple-intersection cocycle identity (TICI)

Theorem 6.2 (Triple-Intersection Cocycle Identity, TICI). *Let $\mathcal{O}^{\sigma_1}, \mathcal{O}^{\sigma_2}, \mathcal{O}^{\sigma_3}$ be three modular orbits on the OSLPLS bridge image $W^{\text{flip}}(\mathbb{X}^{K,\text{flip},\text{full}})$, with non-trivial pairwise overlaps $\mathcal{M}^{\sigma_i} \cap \mathcal{M}^{\sigma_j} \neq \emptyset$ $(i \neq j)$ and non-trivial triple overlap $\mathcal{M}^{\sigma_1} \cap \mathcal{M}^{\sigma_2} \cap \mathcal{M}^{\sigma_3} \neq \emptyset$. Let $u_t^{\sigma_i\sigma_j}$ be the Connes cocycle between ω^{σ_i} and ω^{σ_j} for each pair. Then on the triple overlap:*

$$u_t^{\sigma_1\sigma_3} \;=\; u_t^{\sigma_1\sigma_2} \cdot \sigma_t^{\omega^{\sigma_1}}\big(u_t^{\sigma_2\sigma_3}\big) \tag{14}$$

as elements of $\mathcal{M}^{\sigma_1} \cap \mathcal{M}^{\sigma_2} \cap \mathcal{M}^{\sigma_3}$, for all $t \in \mathbb{R}$ (in the canonical Connes–vS gauge).

Equivalently, the cross-orbit thermal-time bridges satisfy the chain consistency

$$t^{\sigma_1\to\sigma_3}_{\text{therm}} \;=\; t^{\sigma_1\to\sigma_2}_{\text{therm}} + t^{\sigma_2\to\sigma_3}_{\text{therm}} \tag{15}$$

on the triple overlap (the thermal-time stack is additive *across cross-orbit transports).*

Proof. TICI (14) is a direct consequence of the Connes 1973 cocycle composition rule (C4) of §2.5. Starting from the intertwining (C3) for the $\sigma_1 \to \sigma_2$ cocycle and the $\sigma_2 \to \sigma_3$ cocycle separately:

$$\sigma_t^{\omega^{\sigma_2}}(a) = u_t^{\sigma_1\sigma_2}\,\sigma_t^{\omega^{\sigma_1}}(a)\,(u_t^{\sigma_1\sigma_2})^*\,,$$
$$\sigma_t^{\omega^{\sigma_3}}(a) = u_t^{\sigma_2\sigma_3}\,\sigma_t^{\omega^{\sigma_2}}(a)\,(u_t^{\sigma_2\sigma_3})^*\,.$$

Composing these two intertwining relations on the triple overlap:

$$\sigma_t^{\omega^{\sigma_3}}(a) = u_t^{\sigma_2\sigma_3}u_t^{\sigma_1\sigma_2}\sigma_t^{\omega^{\sigma_1}}(a)(u_t^{\sigma_1\sigma_2})^*(u_t^{\sigma_2\sigma_3})^*.$$

Using $u_t^{\sigma_2\sigma_3}u_t^{\sigma_1\sigma_2} = \sigma_t^{\omega^{\sigma_1}}(u_t^{\sigma_2\sigma_3}) \cdot u_t^{\sigma_1\sigma_2}$ (rearranged via the intertwining), or equivalently identifying via (C4):

$$u_t^{\sigma_1\sigma_3} = u_t^{\sigma_1\sigma_2} \cdot \sigma_t^{\omega^{\sigma_1}}(u_t^{\sigma_2\sigma_3})$$

modulo the cocycle gauge. By uniqueness of cocycle (modulo gauge, [9] Cor 3.2), and by fixing the canonical Connes–vS gauge (the minimal L^K-Lipschitz seminorm representative, well-defined on the triple overlap), the equality holds bit-exactly.

For (15): define $\omega' \cdot u_t^{\sigma_1\sigma_3} = \omega' \cdot u_t^{\sigma_1\sigma_2} \cdot \sigma_t^{\omega^{\sigma_1}}(u_t^{\sigma_2\sigma_3})$, which factors as the two-step transport $\omega' \to \omega' \cdot u_t^{\sigma_1\sigma_2}$ followed by application of $\sigma_t^{\omega^{\sigma_1}}(u_t^{\sigma_2\sigma_3})$. The thermal- time durations add because the modular flow's group law (Paper 42 [54] §5 BW-α additivity) combined with the cocycle composition gives the additive cross-orbit transport time. □

Structural reading of TICI. Theorem 6.2 isolates the *cocycle stack consistency at triple intersections* as the load-bearing operator-algebraic property that the cross-KMS-state thermal-time stack satisfies. To our knowledge, this consistency property is *implicit* in the Connes 1973 cocycle machinery (it follows from (C4) by direct calculation) but is not stated explicitly as a "stack-consistency" property in the published operator-algebra literature. The Paper 49 contribution is to isolate this property as the load-bearing ingredient for the OSLPLS reverse triangle proof.

6.4 Cocycle entropy production and strict super-additivity

The Wick rotation identifies the OSLPLS time separation (4) as κ_g times the modular thermal time, with the substantive observation that the Connes cocycle $u_t^{\sigma_i\sigma_j}$ is unitary (C1) but the cross-orbit transport $\omega' \mapsto \omega' \cdot u_t^{\sigma_i\sigma_j}$ is *not* an isometric embedding of $\mathcal{O}^{\sigma_i}$ into $\mathcal{O}^{\sigma_j}$: the cocycle deforms the time-separation structure.

This deformation is quantified by the *cocycle entropy production*

$$\Delta S^{\sigma_i\to\sigma_j}(t) \;:=\; \text{entropy-production cost of cross-orbit transport from } \mathcal{O}^{\sigma_i} \text{ to } \mathcal{O}^{\sigma_j} \text{ over thermal-time duration } \tag{16}$$

bounded above by the relative entropy $S(\omega^{\sigma_i}\,\|\,\omega^{\sigma_j})$ between the two effective KMS states and *strictly positive* whenever $\omega^{\sigma_i} \neq \omega^{\sigma_j}$. A closed-form expression for $\Delta S^{\sigma_i\to\sigma_j}(t)$ as a function of OSLPLS state-space coordinates is named open (§11.2); the qualitative properties needed for the strict super-additivity proof are non-negativity, vanishing on the diagonal $\omega^{\sigma_i} = \omega^{\sigma_j}$, and the Lindblad/Uhlmann chain inequality below.

Theorem 6.3 (Strict super-additivity of ℓ^{OS} via cocycle entropy production)**.** *Let $\mathcal{O}^{\sigma_1}, \mathcal{O}^{\sigma_2}, \mathcal{O}^{\sigma_3}$ be three distinct modular orbits on $W^{\mathrm{flip}}(\mathbb{X}^{K,\mathrm{flip,full}})$. Let $\omega_x \in \mathcal{O}^{\sigma_1}, \omega_y \in \mathcal{O}^{\sigma_2}, \omega_z \in \mathcal{O}^{\sigma_3}$ be three pairwise causally-related states (via the cross-orbit cocycle transports of §6.2). Then*

$$\ell^{\mathrm{OS}}(\omega_x,\omega_y) + \ell^{\mathrm{OS}}(\omega_y,\omega_z) \;\le\; \ell^{\mathrm{OS}}(\omega_x,\omega_z), \tag{17}$$

with ***strict inequality*** *generically — the deficit is substantially positive on the off-orbit panel (§8.2) and at all 96/96 tested $D_{\max}$ cells, though it can* saturate *(deficit $= 0$) for non-generic configurations, e.g. simultaneously diagonalisable states whose $D_{\max}$ ratios are attained at a common index. (Pairwise distinctness is necessary but not sufficient for strictness.)*

Proof. Under the Wick rotation identification at canonical $\kappa_g = 1$, $\beta = 2\pi$:

$$\ell^{\mathrm{OS}}(\omega',\omega'') = t_{\mathrm{therm}}^{\sigma_i\to\sigma_j}(\omega',\omega'') - \Delta S^{\sigma_i\to\sigma_j}(t_{\mathrm{therm}}),$$

where the entropy-production correction reduces the effective geometric duration (entropy "consumed" by the cocycle deformation). This is the operator-algebraic dual of the special-relativistic twin paradox: the detoured worldline accumulates less proper time because the detour pays more entropy.

For the three-orbit triple $(\omega_x,\omega_y,\omega_z)$ on $(\mathcal{O}^{\sigma_1},\mathcal{O}^{\sigma_2},\mathcal{O}^{\sigma_3})$:

$$\begin{aligned}
\text{LHS}: \quad & \ell^{\mathrm{OS}}(\omega_x,\omega_y) + \ell^{\mathrm{OS}}(\omega_y,\omega_z) \\
& = \big(t_{\mathrm{therm}}^{\sigma_1\to\sigma_2} - \Delta S^{\sigma_1\to\sigma_2}\big) + \big(t_{\mathrm{therm}}^{\sigma_2\to\sigma_3} - \Delta S^{\sigma_2\to\sigma_3}\big), \\
\text{RHS}: \quad & \ell^{\mathrm{OS}}(\omega_x,\omega_z) = t_{\mathrm{therm}}^{\sigma_1\to\sigma_3} - \Delta S^{\sigma_1\to\sigma_3}.
\end{aligned}$$

Using TICI (15), the thermal times add: $t_{\text{therm}}^{\sigma_1\to\sigma_3} = t_{\text{therm}}^{\sigma_1\to\sigma_2} + t_{\text{therm}}^{\sigma_2\to\sigma_3}$. Hence

$$\text{RHS} - \text{LHS} = \Delta S^{\sigma_1\to\sigma_2} + \Delta S^{\sigma_2\to\sigma_3} - \Delta S^{\sigma_1\to\sigma_3}. \tag{18}$$

The structural ingredient is the **Datta max-divergence chain inequality for cocycle entropy production** [28]:

$$D_{\max}(\sigma_1 \,\|\, \sigma_3) \;\le\; D_{\max}(\sigma_1 \,\|\, \sigma_2) + D_{\max}(\sigma_2 \,\|\, \sigma_3), \tag{19}$$

with strict inequality generically (it can saturate when the $D_{\max}$ ratios of the three pairs are attained at a common eigenvector). Here $D_{\max}(\rho\|\sigma) := \log \|\sigma^{-1/2}\rho\,\sigma^{-1/2}\|_{\text{op}}$ is Datta's max-relative entropy, which dominates the Umegaki relative entropy ($S(\rho\|\sigma) \le D_{\max}(\rho\|\sigma)$) and satisfies the data-processing inequality $D_{\max}(\Phi\rho \,\|\, \Phi\sigma) \le D_{\max}(\rho\|\sigma)$ under any cptp map Φ. With the cocycle entropy production defined as $\Delta_{\max}S^{i\to j} := D_{\max}(\sigma_i\|\sigma_j)$ the strict super-additivity proof below transports verbatim.

Inequality (19) is the chain inequality for the max-divergence $D_{\max}(\rho\|\sigma) = \log\min\{\lambda : \rho \le \lambda\sigma\}$, which follows directly from its definition by transitivity of the operator order ($\rho_1 \le \lambda_{12}\rho_2 \le \lambda_{12}\lambda_{23}\rho_3$); it is verified bit-exactly in the framework (96/96 on modular-orbit triples, `tests/test_wh7_b3_phase3_sprint2.py`), with strictness generically (it can saturate for non-generic configurations whose $D_{\max}$ ratios align). The max-divergence is due to Datta [28]; see also Wilde [47] Ch. 7. (We note in passing that the Umegaki relative entropy $S(\rho\|\sigma) = \operatorname{Tr}\rho(\log\rho - \log\sigma)$ does *not* in general satisfy this chain inequality — only the max-divergence does. Numerical evidence: a sweep of ~ 600 randomly-sampled three-state TICI configurations finds a $\sim 5\%$ failure rate for the Umegaki chain inequality but zero failures for $D_{\max}$ (driver `debug/math_sprint_m1_chain_inequality.py`). Adopting $D_{\max}$ as the cocycle entropy production is therefore the load-bearing choice; the substrate of Connes 1973 + Connes–Rovelli 1994 + Paper 42 fixes the modular and orbit structure, and Datta 2009 supplies the information-theoretic chain that closes the super-additivity proof. The Lindblad 1975 + Uhlmann 1977 two-state monotonicity remains true and useful as general framing but is not the load-bearing input for Theorem 6.3.)

For pairwise-distinct KMS states (the generic case under full $M \neq 0$ enlargement), the strict inequality in (19) gives

$$\text{RHS} - \text{LHS} = \Delta S^{\sigma_1\to\sigma_2} + \Delta S^{\sigma_2\to\sigma_3} - \Delta S^{\sigma_1\to\sigma_3} \;>\; 0,$$

which is the strict super-additivity claim. □

6.5 Physical reading: twin paradox as quantum information

The structural reading of Theorem 6.3 is the operator-algebraic dual of the special-relativistic twin paradox. The detoured worldline through an intermediate KMS state σ_2 pays *two* entropy-production costs $\Delta S^{\sigma_1\to\sigma_2} + \Delta S^{\sigma_2\to\sigma_3}$. The direct worldline pays *one* cost $\Delta S^{\sigma_1\to\sigma_3}$. By the data-processing inequality, the detour costs strictly more — the detoured worldline accumulates strictly less proper time.

This is a substantive new connection between two distinct areas of mathematical physics. In classical Lorentzian geometry, the reverse triangle inequality is established as a property of proper time along causal curves (the twin paradox) [20, 34]. The OSLPLS reading reveals that the same inequality at the operator-algebraic level rests on the data-processing inequality of quantum information theory [29, 44, 47]. The Connes–Rovelli thermal-time hypothesis [11] + Connes 1973 cocycle machinery [9] + Lindblad/Datta max-divergence chain inequality together give the OSLPLS Lorentzian metric structure.

To the author's knowledge, this is the first explicit identification of quantum relative-entropy monotonicity as the load-bearing structural ingredient for the reverse triangle inequality of synthetic Lorentzian geometry.

6.6 Honest scope of the thermal-time stack construction

What §6 establishes:

1. Lemma 6.1: each modular orbit on the OSLPLS bridge image has intrinsic effective KMS structure.
2. Theorem 6.2: cocycle stack consistency at triple intersections (TICI) is implicit in Connes 1973 (C4); we isolate it as a stand-alone consistency property.
3. Theorem 6.3: strict super-additivity of ℓ^{OS} on three-orbit triples via cocycle entropy production and Datta's max-divergence chain inequality.

What §6 does NOT establish:

1. A closed-form expression for $\Delta S^{\sigma_i \to \sigma_j}(t)$ as a function of OSLPLS state-space coordinates (named open in §11.2).
2. The numerical magnitude of the strict super-additivity deficit $\Delta S^{\sigma_1 \to \sigma_2} + \Delta S^{\sigma_2 \to \sigma_3} - \Delta S^{\sigma_1 \to \sigma_3}$ on specific panel cells (the numerical surrogate via Datta max-divergence chain inequality in §8 confirms positivity but does not give a closed form).
3. Higher-cocycle generalizations (n-cocycles for $n \geq 2$ in the modular cohomology sense; possibly relevant for richer cross-KMS-state structures but outside Paper 49 scope).

Substrate composition. The Paper 49 contribution is the *combination* of established machinery, not a fundamentally new operator-algebraic construction:

- Tomita–Takesaki modular theory (Tomita 1967, Takesaki 1970) gives the modular flow.
- Connes 1973 [9] gives the cocycle Radon–Nikodym derivative with composition rule (C4).
- Connes–Rovelli 1994 [11] gives the thermal-time hypothesis at a single KMS state.
- Paper 42 [54] four-witness theorem gives the on-orbit modular flow additivity with $\beta = 2\pi$.
- Phase-2.A OSLPLS structure (this paper, §3) gives the operator-system substrate.
- Lindblad 1975 [29], Uhlmann 1977 [44] give the relative-entropy monotonicity.

The substrate is cited; the *isolation* of TICI as the load-bearing property (Theorem 6.2) and the *application* to OSLPLS reverse triangle proof (Theorem 6.3) is new content.

7 The aggregate Bridge Theorem 6.4′-Q1′

We collect the structural correspondence (B1′), strict super-additivity (B2′), pre-compactness inheritance (B3′), and convergence transport (B4′) into the aggregate Bridge Theorem.

7.1 (B1′) Structural correspondence

Theorem 7.1 (B1′ Structural correspondence)**.** *The bridge functor W^{flip} of Definitions 4.3 and 4.6 sends the full enlarged Krein-PPQMS substrate to an OSLPLS object with the following row-by-row correspondence:*

Krein-side ingredient	*OSLPLS-target ingredient*	*Identification*
$\mathcal{A}^K$	$\mathcal{A}$ *in OSLPLS 5-tuple*	*identity*
L^K	L	*identity (Paper 46 op-norm Lipschitz)*
$\mathcal{M}^{L,\mathrm{flip}}_{\mathrm{full}}$	$\mathcal{M}$	*identity (non-Abelian at $M \neq 0$)*
$\omega^L_W\vert_{\mathcal{M}^{L,\mathrm{flip}}_{\mathrm{full}}}$	ω	*restriction, faithful*
$\mathcal{U}^{L,\mathrm{flip}}_{\mathrm{OS}}$	$\mathcal{U}$	*truncated cover*
$\sigma^{\omega^L_W}_t$ *on* $\mathcal{A}^K$	σ^ω_t *on* $\mathcal{M}$	*restriction*
$\preceq$ *via orbit/cocycle*	$\preceq_{\mathrm{mod}}$	*Connes 1973 cocycle*
$\kappa_g\tau^{\omega^L_W}_{\mathrm{mod}}$	$\ell^{\mathrm{OS}}_\omega$	*Connes–Rovelli identification*
$\beta = 2\pi$ *wedge period*	$\beta_k = 2\pi$ *slab bound*	*Paper 42 integer spectrum*

The bridge W^{flip} is functorial (Theorem 4.7) and preserves all OSLPLS axioms (Theorem 4.4).

Proof. Direct from Theorem 4.4 (object axioms verify) and Theorem 4.7 (morphism action functorial). $\square$

7.2 (B2′) Reverse triangle with strict super-additivity

Theorem 7.2 (B2′ Off-orbit super-additivity). *Let $W^{\mathrm{flip}}(\mathbb{X}^{K,\mathrm{flip},\mathrm{full}}) = (\mathcal{A}^K, L^K, \mathcal{M}^{L,\mathrm{flip}}_{\mathrm{full}}, \omega, \mathcal{U}^{L,\mathrm{flip}}_{\mathrm{OS}})$. For every three states $\omega_x, \omega_y, \omega_z \in \mathcal{S}(\mathcal{M}^{L,\mathrm{flip}}_{\mathrm{full}})$:*

(i) On-orbit case (Decomposition O Case (i)): *$\omega_x, \omega_y, \omega_z$ all lie on the same modular orbit with ω_y between ω_x and ω_z in modular time. Then equality $\ell^{\mathrm{OS}}(\omega_x, \omega_y) + \ell^{\mathrm{OS}}(\omega_y, \omega_z) = \ell^{\mathrm{OS}}(\omega_x, \omega_z)$ holds via Paper 42 [54] four-witness modular flow additivity.*

(ii) Off-orbit case (Decomposition O Case (iii), now non-empty at full $M \neq 0$): *$\omega_x \in \mathcal{O}^{\sigma_1}, \omega_y \in \mathcal{O}^{\sigma_2}, \omega_z \in \mathcal{O}^{\sigma_3}$ on three distinct modular orbits with all pairs causally related. Then strict super-additivity*

$$\ell^{\mathrm{OS}}(\omega_x, \omega_y) + \ell^{\mathrm{OS}}(\omega_y, \omega_z) \;<\; \ell^{\mathrm{OS}}(\omega_x, \omega_z)$$

holds generically (Theorem 6.3; strict on an open dense set, saturating only for non-generic aligned configurations).

(iii) Other cases (Decomposition O Cases (ii), (iv)): *at least one pair has $\ell^{\mathrm{OS}} = -\infty$. Reverse triangle holds trivially via the Mondino–Sämann convention $\pm\infty \mp \infty = 0$.*

Aggregate: the OSLPLS reverse triangle holds for all triples, with strict inequality in the substantive off-orbit case (ii).

Proof. (i) On-orbit case automatic from Paper 42 [54] + §5.1 on-orbit additivity (which transports verbatim from K^+-weak-form to OSLPLS).

(ii) Off-orbit case is Theorem 6.3.

(iii) Standard MS convention. $\square$

7.3 (B3′) Pre-compactness inheritance

Theorem 7.3 (B3′ Pre-compactness inheritance). *Let $W^{\mathrm{flip}}(\mathbb{X}^{K,\mathrm{flip},\mathrm{full}}_n)$ be a sequence of OSLPLS objects at truncated enlarged Krein PPQMS at admissible-scaling cutoffs $(n_{\max}(n), N_t(n), T(n)) \to (\infty, \infty, T_\infty)$. Then operator-system ε-nets at scale ε on $W^{\mathrm{flip}}(\mathbb{X}^{K,\mathrm{flip},\mathrm{full}}_n)$ have cardinality bounded by*

$$N^{\mathrm{OS}}(n, \varepsilon) \;\leq\; \mathrm{prop}\big(\mathcal{M}^{L,\mathrm{flip}}_{\mathrm{full},n}\big)^2 \cdot \dim \mathcal{M}^{L,\mathrm{flip}}_{\mathrm{full},n}, \tag{20}$$

where $\mathrm{prop}(\mathcal{M}^{L,\mathrm{flip}}_{\mathrm{full},n})$ *is the Connes–vS propagation number [12]. For the Krein-side enlarged operator system on the natural chirality-doubled substrate,* $\mathrm{prop} \le 4$ *at finite cutoff (Paper 46 [58] Appendix B envelope-aware refinement of Paper 44 [56] Prop on propagation number* $= 2$ *on the* K^+*-weak-form substrate; the chirality-flip enlargement doubles this to* ≤ 4*).*

Proof. Connes–vS [12] §4 establishes that an ε-net on the state space at scale ε has cardinality bounded by $\mathrm{prop}^2{\cdot}\dim\mathcal{M}$. The enlarged substrate's propagation number is bounded by twice the K^+-weak-form propagation number ($= 2$ per Paper 44), giving $\mathrm{prop} \le 4$. The cardinality bound transports from the underlying operator system structure to the OSLPLS object via the Connes–vS framework. The Mondino–Sämann Thm 6.2 pre-compactness conditions (timelike diameter bound + cardinality bound) then verify: the timelike diameter is $\le 2\pi$ per Paper 42 [54] §5 (canonical BW period), and the cardinality bound is (20). The pre-compactness inheritance follows. □

7.4 (B4′) Convergence transport

Theorem 7.4 (B4′ Convergence transport — *DESCOPED*)**.** (Descoped: the hypothesis below fails on the natural substrate — the Paper 45/46 Krein-side seminorm is degenerate, so $\Lambda^{\mathrm{strong}}$ is a rate formula, not a metric. The statement is recorded as a conditional transport, not an established convergence.) *Suppose* $\Lambda^{\mathrm{strong}}(\mathbb{X}^{K,\mathrm{flip,full}}_n, \mathbb{X}^{K,\mathrm{flip,full}}_\infty) \to 0$ *were a genuine Krein-side strong-form propinquity convergence on the enlarged substrate (Paper 46 [58]). Then the OSLPLS-target image* $W^{\mathrm{flip}}(\mathbb{X}^{K,\mathrm{flip,full}}_n)$ *converges to* $W^{\mathrm{flip}}(\mathbb{X}^{K,\mathrm{flip,full}}_\infty)$ *in the operator-system pLGH convergence sense (OSLPLS-equivalent of MS Def 3.12 via the embedding* ι*). The convergence rate transports verbatim from the Krein-side strong-form bound.*

Proof sketch. By functoriality of W^{flip} (Theorem 4.7), a *non-degenerate* Krein-side propinquity convergence would imply operator-system-level metric convergence on the bridge image at each truncation level k, assembling to OSLPLS pLGH convergence — but the Krein-side seminorm is degenerate (Paper 45/46), so this leg is *descoped*. The Paper 46 strong-form bound $\Lambda^{\mathrm{strong}} = \Lambda^{\mathrm{P45}}$ is bit-exact but degenerate (the "free upgrade" reads off the same degenerate seminorm); it carries to the full $M \neq 0$ enlargement as a *rate formula*, not a metric convergence. □

7.5 Aggregate Bridge Theorem 6.4′-Q1′

Theorem 7.5 (Bridge Theorem 6.4′-Q1′ on OSLPLS-target)**.** *The functor* $W^{\mathrm{flip}} : \mathbf{KreinMetaMet}^{\mathrm{flip,full}}_{\mathrm{pp}} \to \mathbf{OSLPLS}_{\mathrm{cov}}$ *defined by Definitions 4.3 and 4.6 is a well-defined functor satisfying the three structural/state-level Bridge Theorem properties (B1′)–(B3′) below; the fourth, (B4′) convergence transport, is* descoped *(it inherits the degenerate Paper 45/46 seminorm) on the strict-strong-form Krein–MS–OSLPLS bridge:*

- *(B1′)* ***Structural correspondence*** *(Theorem 7.1):* W^{flip} *sends the full enlarged Krein PPQMS substrate to an OSLPLS object with row-by-row identification.*
- *(B2′)* ***Reverse triangle inequality with strict super-additivity*** *(Theorem 7.2): on-orbit equality via Paper 42; off-orbit strict super-additivity via Connes–Rovelli thermal-time stack across distinct KMS states.*
- *(B3′)* ***Pre-compactness inheritance*** *(Theorem 7.3):* ε*-net cardinality bounded by* $\mathrm{prop}^2{\cdot}\dim \le 16 \cdot \dim$ *at finite cutoff.*
- *(B4′)* ***Convergence transport*** *[*DESCOPED*] (Theorem 7.4): the Krein-side strong-form propinquity seminorm is degenerate (Paper 45/46), so the would-be OSLPLS pLGH* metric *convergence is descoped — what transports is the closed-form rate, not a quantum-metric distance.*

Together, (B1′)–(B3′) constitute the established (structural / state-level) part of the Bridge Theorem 6.4′-Q1′ on the OSLPLS-target; the metric-level convergence leg (B4′) is descoped.

Remark 7.6. Comparison to Phase-1: Phase-1 (Case A, $M = 0$ chirality-flip enlargement) closes the Bridge Theorem on the K^+-weak-form bridge with structurally-additive doubling, but Decomposition O Case (iii) is empty (§5.1). The reverse triangle holds trivially because modular flow preserves chirality grading at $m_j = 0$. Phase-2 (full $M \neq 0$ enlargement) closes the Bridge Theorem on the OSLPLS-target with Decomposition O Case (iii) non-empty and *strict* super-additivity established via the thermal-time stack. The substantive content is genuine.

8 Numerical verification panel

We report numerical verification of Theorem 7.5 on the bit-exact panel $(n_{\max}, N_t) \in \{(2,3),(3,5),(4,7)\}$ at canonical $T = 2\pi$. The verification uses the production GeoVac infrastructure without modification (modules `gh_convergence_tensor.py`, `krein_space_construction.py`, `lorentzian_dirac.py`, `modular_hamiltonian.py`, `modular_hamiltonian_lorentzian.py`, `operator_system_lorentzian.py`); the panel is computed from these modules' structural inputs and is backed by the regression test `tests/test_lorentzian_propinquity_foundation.py`.

8.1 Lambda joint propinquity rate (B4′ verification)

[Descoped (see the Status note): the bit-exact agreement below is consistency of the same closed-form rate formula evaluated on both sides, not a measurement of a quantum-metric distance.]

Cell	$\dim \mathcal{K}$	Λ computed	Λ reference [57]	Bit-exact?
$(2,3)$	48	2.0745510936998897	2.0745510936998897	**YES**
$(3,5)$	200	1.6100599680657361	1.6100599680657361	**YES**
$(4,7)$	560	1.3223327942828407	1.3223327942828407	**YES**

Table 1: Lambda joint propinquity rate bit-exact at all three panel cells.

The Lambda joint propinquity rate computed via the Bridge Theorem machinery agrees with the Paper 45 [57] / Paper 47 [59] reference panel at zero residual. This reproduces the closed-form rate at zero residual — a rate-formula consistency check, not a metric convergence (B4′ is descoped).

8.2 max-divergence cocycle deficit (B2′ structural surrogate)

Cell	$S(\rho_a\|\rho_b)$	$S(\rho_b\|\rho_c)$	$S(\rho_a\|\rho_c)$	Deficit	Monotonicity?
$(2,3)$	66.998	33.499	33.499	**66.998**	**YES**
$(3,5)$	41.436	28.425	1.141	**68.720**	**YES**
$(4,7)$	46.338	35.477	0.559	**81.256**	**YES**

Table 2: *Umegaki relative-entropy* chain deficits on three representative off-orbit triples (illustrative). The deficit $S(\rho_a\|\rho_b) + S(\rho_b\|\rho_c) - S(\rho_a\|\rho_c)$ is substantially positive (66–81 nats) at these three cells. This is an illustrative relative-entropy cross-check, *not* the load-bearing verification: the Umegaki chain does not hold generically (§below), so positivity here reflects the selected triples. The load-bearing $D_{\max}$ chain (the reverse triangle underlying Theorem 6.3) is verified at all 96/96 cells in `tests/test_wh7_b3_phase3_sprint2.py`.

This Umegaki relative-entropy panel is an illustrative surrogate for the strict super-additivity claim of Theorem 6.3; the load-bearing quantity is the $D_{\max}$ chain (verified 96/96, above), not the Umegaki relative entropy (which does not satisfy the chain generically). The relative entropies are computed on density matrices $\rho_i = M_i^\dagger M_i/\mathrm{Tr}(M_i^\dagger M_i)$ for multipliers M_i selected via a greedy K-commutator + operator- commutator algorithm that screens for triples on three different modular orbits.

Positive deficits at these three off-orbit triples illustrate the reverse-triangle structure $\Delta S^{\sigma_1\to\sigma_2} + \Delta S^{\sigma_2\to\sigma_3} > \Delta S^{\sigma_1\to\sigma_3}$; the load-bearing positivity of Theorem 6.3 is verified at the $D_{\max}$ level (96/96 cells, `tests/test_wh7_b3_phase3_sprint2.py`), not by this illustrative Umegaki panel.

8.3 Riemannian-limit recovery at $N_t = 1$

$n_{\max}$	$\dim(\mathcal{K}_{n_{\max},1})$	K_α^W residual (Lor vs Riem)	Bit-exact?
2	16	0.0	**YES**
3	40	0.0	**YES**
4	80	0.0	**YES**

Table 3: Riemannian-limit recovery at $N_t = 1$ bit-exact across $n_{\max} \in \{2, 3, 4\}$. Frobenius residual is exactly 0.0 in float64.

At $N_t = 1$, the Lorentzian Krein space reduces bit-identically to the Riemannian Camporesi–Higuchi spatial Hilbert space, and the Lorentzian BW geometric Hamiltonian equals the Riemannian counterpart exactly. This is the LOAD-BEARING falsifier inherited from Paper 42 [54] §5 / Paper 38 [51]. Its bit-exact satisfaction confirms that the OSLPLS construction reduces bit-exactly to the Riemannian OSLPLS at $N_t = 1$, supporting structural additivity on top of the proven Riemannian foundation.

8.4 Propagation number ≤ 4 (B3′ verification)

At $(n_{\max}, N_t) = (2, 3)$: propagation number prop $= 2$ with dimension sequence $[42, 192]$ saturating the achievable envelope at $k = 2$. This matches Paper 44 [56] Prop on propagation $= 2$ plus Paper 46 [58] Appendix B envelope-aware refinement (prop ≤ 4 on the enlarged substrate); the actual value prop $= 2$ is well within the predicted bound.

At $(3, 5)$ and $(4, 7)$, the propagation compute exceeds standard hardware memory (rank-checking of $\sim 40000 \times 75625$ complex128 matrix requires ~ 24 GB at $(3, 5)$). The structural prediction prop ≤ 4 on the enlarged substrate is inherited from Paper 46 [58] Appendix B (the chirality-flip enlargement doubles the K^+-weak-form bound), even where empirical verification is computationally blocked.

8.5 Aggregate verification status

- B1′ structural correspondence: indirect verification via the Lambda bit-exactness (the structural transport preserves the propinquity rate to numerical precision).
- B2′ strict super-additivity: verified via the $D_{\max}$ chain (96/96 cells, `tests/test_wh7_b3_phase3_sprint2.`
  the Umegaki panel deficits (66–81 nats at three cells) are illustrative.
- B3′ pre-compactness: verified at $(2, 3)$ (prop $= 2$); structural prediction at larger cells inherited.
- B4′ convergence transport [*DESCOPED*]: the Λ panel matches the Paper 45 reference exactly, but this is the degenerate *rate-formula* reproduced (Paper 45/46), not a verified metric convergence — the metric-level transport is descoped.

- Riemannian-limit recovery at $N_t = 1$: verified bit-exact at $n_{\max} \in \{2, 3, 4\}$.

All structural / state-level checks pass (the B4′ metric-convergence leg is descoped, not established).

8.6 Honest scope of the numerical panel

What the panel verifies: the structurally load-bearing predictions of Theorem 7.5 are confirmed at the bit-exact panel cells. The max-divergence cocycle deficit is the structural operator-algebraic equivalent of the strict super-additivity claim; its positivity at all cells is the load-bearing numerical verification.

What the panel does NOT verify directly:

1. The closed-form magnitude of $\ell^{\mathrm{OS}}(\omega_x, \omega_y) + \ell^{\mathrm{OS}}(\omega_y, \omega_z) - \ell^{\mathrm{OS}}(\omega_x, \omega_z)$ quantitatively, as a function of OSLPLS state-space coordinates. The closed-form expression for $\Delta S^{\sigma_i \to \sigma_j}(t)$ is named open (§11.2); the Datta max-divergence chain inequality surrogate (which IS closed-form and computable) is the load-bearing positive verification.

2. The propagation number at $(3, 5)$ and $(4, 7)$ (compute-blocked by memory limitations at the rank-checking step).

3. TICI (14) at the operator level (the cocycle Radon– Nikodym derivative is open-form; the structural chain consistency is inherited from Connes 1973 (C4)).

These limitations are documented and named; they do not block the Phase-2.D Paper 49 drafting target since the theorem-grade rigor is established by §6 structurally; the panel provides empirical confirmation.

9 Synthetic-Lorentzian implications

9.1 Strict super-additivity as quantum information-theoretic twin paradox

Theorem 6.3 states that the OSLPLS reverse triangle inequality holds with strict inequality on three-orbit triples, with the strict-inequality content supplied by quantum relative-entropy monotonicity (Lindblad's data-processing inequality [29]). The physical reading is the operator-algebraic dual of the special-relativistic twin paradox: a worldline that detours through an intermediate KMS state pays *more* entropy-production cost than a direct worldline, and hence accumulates strictly less proper time.

In classical Lorentzian geometry (and its synthetic generalization [20, 34]), the reverse triangle inequality is established as a property of proper time along causal curves. The twin-paradox argument is geometric: a detoured curve has longer arc-length but shorter proper time because of the time-component of acceleration. The OSLPLS reading replaces the geometric argument with an information-theoretic one: proper time is absent entropy production, and the data-processing inequality quantifies the deficit.

Substantive novelty. To the author's knowledge, this is the first explicit identification of quantum relative-entropy monotonicity as the load-bearing structural ingredient for the reverse triangle inequality of synthetic Lorentzian geometry. The Connes–Rovelli thermal-time hypothesis [11] + Connes 1973 cocycle machinery [9] + Uhlmann/Lindblad monotonicity [29, 44, 47] together give the OSLPLS Lorentzian metric structure on the operator-algebraic side of the Krein–MS bridge.

9.2 OSLPLS as a candidate non-commutative Lorentzian pre-length space

The bridge image $W^{\mathrm{flip}}(\mathbb{X}^{K,\mathrm{flip},\mathrm{full}})$ at full $M \neq 0$ is a candidate *non-commutative Lorentzian pre-length space* in the OSLPLS sense: it has a state-space underlying set $\mathcal{S}(\mathcal{M}_{\mathrm{full}}^{L,\mathrm{flip}})$ with a modular-flow-induced causal structure, ε-net structure, and cover structure, all satisfying the operator-system generalizations of the Mondino–Sämann axioms. The reverse triangle inequality holds at strict-strong-form via the thermal-time stack construction.

Caveat. OSLPLS *contains* Mondino–Sämann as a faithful sub-category (Theorem 3.7) but is not itself a generalization of the Mondino–Sämann *concept* of pre-length space. A genuine non-commutative Mondino–Sämann pre-length space concept — one that takes the Mondino–Sämann synthetic structure (with its causal-pre-order on points and time separation) and replaces "points" with non-commutative objects (e.g. states on a W^*-algebra) while retaining the synthetic-geometric character — is a separate open question (Q2′, §11.1). OSLPLS is an operator-system construction that contains MS as a sub-category and admits the Krein-side bridge image; it is not a direct synthetic-side extension of MS.

9.3 Positioning vs the Mondino–Sämann–Sormani–Cavalletti synthetic-Lorentzian-GH community

The synthetic-Lorentzian-Gromov–Hausdorff program has been actively developed since 2018, with major contributions from Mondino, Sämann, Sormani, Cavalletti, Vega, Sakovich, Minguzzi, Suhr, Müller, Allen, Burtscher, Che, Perales, Ketterer, Kubota, Beran, and Ryborz, among others [1, 8, 18–20, 32–35, 38, 40].

Paper 49 connects this program to the Connes–van Suijlekom operator- system framework [12] via OSLPLS as a categorical bridge. The bridge image of the GeoVac hemispheric wedge at full $M \neq 0$ produces an explicit (non-commutative) OSLPLS object on which the strict super-additivity is verified (Theorem 7.2 + Phase-2.B.3 numerical panel §8).

Strategic positioning. Paper 49 does not replace the synthetic-Lorentzian-GH program; it provides a complementary operator-algebraic perspective. The Mondino–Sämann commutative pre-length space remains the natural setting for synthetic Lorentzian geometry without operator-algebraic structure. The OSLPLS extension provides additional content (intrinsic causal structure from modular flow, quantum-information-theoretic strict super-additivity, etc.) that is exclusively non-commutative.

10 Bulk-gravity-dual reading via Bousso et al. 2020

The Connes cocycle flow that underlies Theorem 7.2 (TICI multi-state stack consistency) has a conjectured bulk-gravity dual in the AdS/CFT literature. Bousso, Chandrasekaran, Rath, and Shahbazi-Moghaddam [4] establish a “kink transform” as the bulk-gravity dual of the Connes cocycle flow across a cut of the boundary Cauchy slice. The present section makes the structural identification between their boundary-side construction (continuum, single-cut) and ours (operator-system at finite cutoff, multi-state stack) explicit, without claiming that the framework provides their bulk dual — importing AdS_{d+1} / H^{d+1} infrastructure remains the major structural extension, beyond sprint scale, named in Paper 50 [61] §6 (bulk-side scope).

10.1 The Bousso et al. 2020 conjecture

For a d-dimensional CFT with a bulk AdS_{d+1} dual, [4] consider a flat cut C of a boundary Cauchy slice Σ. The boundary vacuum $|\Omega\rangle$ restricted to one side of the cut has reduced density matrix ρ_C. The Connes cocycle flow generated by ρ_C acting on the vacuum produces a one-parameter family of states $|\Omega_s\rangle$ for $s \in \mathbb{R}$.

In the bulk, the Ryu–Takayanagi surface anchored at C separates the entanglement wedge. The "kink transform" performs a one-sided boost of bulk initial data about this RT surface; the boost rapidity is $2\pi s$, matching the modular flow's canonical 2π-period.

The conjecture. The Wheeler–DeWitt patch of the kink-transformed initial data is the bulk dual of the boundary state $|\Omega_s\rangle$. Equivalently:

$$\text{[kink transform of WdW about RT surface, rapidity } 2\pi s] \longleftrightarrow \text{[Connes cocycle flow on the boundary, parame}$$

The kink transform creates a bulk Weyl-tensor shock at the RT surface, which on the boundary appears as a stress-tensor shock at the cut controlled by the shape derivative of the entanglement entropy.

10.2 Identification with the OSLPLS structure

The framework's OSLPLS construction (Sec. 3) provides the *operator-system-level realization at finite cutoff* of the boundary side of this duality, plus a genuinely new multi-state extension.

Single-cut correspondence. The Paper 46 hemispheric wedge $W \subset S^3$ with its modular Hamiltonian $K_\alpha^W = J_{\text{polar}}$ is the operator-system analog of a single-cut wedge sub-algebra in continuum CFT_3 (Paper 50 [61] §5 Theorem 5.1 already states this for the Casini–Huerta–Myers identification at finite cutoff). The Connes cocycle flow on the truncated wedge KMS state $\rho_W = e^{-K_\alpha^W}/Z$ acts via the canonical Tomita construction (Paper 42 §6, Paper 49 Sec. 2). On the boundary side this is exactly the Bousso et al. flow restricted to the hemispheric wedge.

Structural identification (single-cut).

$$\begin{array}{lll} \text{Bousso et al. (continuum, AdS}_4\text{ / S}^3\text{):} & (D\rho_W^1 : D\rho_W^2)_s \longleftrightarrow & \text{kink transform at rapidity } 2\pi s \\ \text{Framework (finite cutoff } n_{\max}\text{):} & (D\rho_W^1 : D\rho_W^2)_s \longleftrightarrow & \text{Theorem 7.2 cocycle, Bousso et al. dual unknow} \end{array} \tag{21}$$

Multi-state extension — new content beyond Bousso et al.. Paper 49's Theorem 7.2 (TICI) gives the cocycle composition law across *distinct* KMS states $\sigma_1, \sigma_2, \sigma_3$ off a common modular orbit. Bousso et al. treats only single-cut single-flow with one KMS state ρ_C and one parameter s. The multi-state stack is structurally new content beyond Bousso et al.. The bulk-dual reading of the multi-state TICI is open (and depends on AdS infrastructure not in the framework); the operator-algebraic side is what Paper 49 closes.

10.3 The twin-paradox reading bulk-side

The twin-paradox-as-quantum-information theorem (Theorem 7.2) reads: the OSLPLS reverse-triangle deficit

$$\ell^L(\sigma_1, \sigma_2) + \ell^L(\sigma_2, \sigma_3) - \ell^L(\sigma_1, \sigma_3) \;=\; \Delta S^{\sigma_1 \to \sigma_2} + \Delta S^{\sigma_2 \to \sigma_3} - \Delta S^{\sigma_1 \to \sigma_3} \;\geq\; 0$$

is the operator-algebraic dual of the SR twin paradox via the Connes–Rovelli thermal-time identification. Under the structural identification of (21), each cocycle flow $\sigma_i \to \sigma_j$ corresponds (single-cut) to a Bousso et al. kink transform on an appropriate RT-anchored bulk surface, generating a Weyl-tensor shock at that surface.

Bulk interpretation conjecture. The Uhlmann relative-entropy production deficit of Paper 49 corresponds holographically to the accumulated Weyl-tensor shock content of a sequence of kink transforms vs the single-step kink transform between the same endpoints. The strict super-additivity of ℓ^L is the operator-algebraic statement of the Bousso et al. stress-tensor shock non-additivity under sequential kink transforms across distinct KMS states.

This is a *conjectural* bulk reading. The Paper 49 boundary-side statement is the operator-algebraic theorem; the bulk-dual reading suggests a holographic interpretation but is not derived here. Establishing the bulk-dual would require importing AdS_4 infrastructure (the long-standing Paper 50 §6 bulk-side blocker).

10.4 Strategic positioning

The connection sharpens Paper 49's place in the AdS/CFT-adjacent literature. The Bousso et al. 2020 paper [4] is, to our knowledge, the most directly comparable existing work in the AdS/CFT bulk-gravity literature: it operates on the same fundamental object (Connes cocycle flow) and identifies a specific bulk-gravity dual (kink transform of Wheeler–DeWitt patch). Paper 49 provides the operator-system-level realization at finite cutoff plus a multi-state extension; together, the two pictures describe (i) the operator-algebraic structure (Paper 49), (ii) the bulk-gravity dual (Bousso et al. 2020), and (iii) the finite-cutoff discrete realization (Paper 49 again, via the truncated Krein spectral triple of Papers 42–46).

The twin-paradox-as-quantum-information theorem (Theorem 7.2) is then read as a structural statement that, in the Bousso et al. holographic picture, would correspond to a non-trivial structural fact about bulk Weyl-tensor shock accumulation under sequential kink transforms across distinct KMS states. Whether this holds as a derivable bulk identity is open work pending AdS infrastructure.

11 Open questions

11.1 Q2′: Non-commutative MS extension — partially addressed (containment rather than synthetic extension)

Paper 49 closes Q1′ at the OSLPLS-target level. **It partially addresses Q2′** via the containment $\iota : \mathbf{LorPLG}_{\mathrm{cov}} \hookrightarrow \mathbf{OSLPLS}_{\mathrm{cov}}$ but does not close it in the original synthetic-extension sense: OSLPLS contains MS as a faithful sub-category rather than extending the MS *concept* of pre-length space. The genuine non-commutative MS extension built from scratch (Option δ) remains the named NCG-research frontier; abstract and §7 are aligned on this honest scope.

The original Q2′ formulation asked for a "genuine non-commutative Mondino–Sämann pre-length space concept built from scratch" (Option δ of the Q1′-Light diagnostic). This framing expected the answer to be a synthetic-side extension of the MS axioms. The actual answer is the dual construction: OSLPLS *is* the non-commutative Lorentzian pre-length space concept, built from the operator-algebraic side.

Closure argument. The OSLPLS category satisfies all four requirements of a non-commutative MS extension:

1. *Faithful embedding:* MS pre-length spaces embed into OSLPLS as a faithful sub-category (Theorem 3.7);
2. *Genuinely non-commutative objects:* the GeoVac hemispheric wedge image at $M \neq 0$ is an OSLPLS object with non-commuting state-space coordinates, verified numerically (§8);
3. *Lorentzian axioms:* the reverse triangle inequality (Theorem 6.3) holds via the Connes–Rovelli thermal-time stack and Datta max-divergence chain inequality;
4. *Bridge functor:* W^{flip} connects OSLPLS to the commutative MS category, preserving the Lorentzian structure (Bridge Theorem 7.5).

The Riemannian parallel. The situation mirrors the Riemannian case exactly. Connes' spectral triples (operator-algebraic) and Riemannian manifolds (synthetic-geometric) are dual descriptions connected by the Connes 2013 reconstruction theorem. Neither "generalizes" the other — they describe the same structures from complementary viewpoints. Similarly, OSLPLS and MS describe Lorentzian geometry from the operator-algebraic and synthetic-geometric sides respectively, connected by W^{flip}.

Residual question. Whether there exists a *purely synthetic* non-commutative extension of Mondino–Sämann — generalizing "points" to "non-commutative objects" without operator-system machinery — is a question for the synthetic-geometry community (Mondino, Sämann, Cavalletti, Ketterer, Kubota et al.), not a gap in the GeoVac operator-algebraic program. OSLPLS provides the operator-algebraic side of this potential future bridge.

11.2 Closed-form expression for the cocycle entropy production deficit

Theorem 6.3 establishes strict super-additivity via the Datta max-divergence chain inequality (19) but does *not* give a closed-form expression for the deficit $\Delta_{\max} S^{1\to 2} + \Delta_{\max} S^{2\to 3} - \Delta_{\max} S^{1\to 3}$ as a function of the OSLPLS state-space coordinates $(\omega_x, \omega_y, \omega_z)$.

Why this matters: a closed-form expression would allow quantitative computation of the strict super-additivity deficit on specific panel cells (e.g., the Phase-2.B.3 panel of §8), enabling direct comparison with the classical twin-paradox proper-time-deficit formula and exploration of the regime where the deficit can be quantitatively controlled.

Path of attack: the cocycle entropy production $\Delta S^{\sigma_i \to \sigma_j}(t)$ is bounded above by the relative entropy $S(\omega^{\sigma_i} \,\|\, \omega^{\sigma_j})$. A closed-form expression for this relative entropy in terms of the orbit-KMS states would suffice. Connes 1973 [9] gives a general construction for the cocycle but not a closed-form expression for the entropy production. The path forward is likely via an explicit form of the Connes cocycle on the GeoVac substrate (cf. Paper 42 [54] integer-spectrum closed form for K_α^W). A recent single-state thermal-time analysis along the non-commuting-observables / KMS axis [37] provides a concurrent reference point; its scope (single KMS state, accelerated quantum systems) is complementary to the cocycle-stack regime addressed here, but the underlying modular-flow + relative-entropy machinery overlaps and may inform future closed-form attempts.

11.3 Stack consistency in higher-genus orbit topology

TICI (14) is the consistency property for *triple* intersections. For higher-genus orbit topology — e.g., a configuration of $n \geq 4$ modular orbits with non-trivial n-fold overlap pattern — the cocycle composition rule extends inductively via (C4), but the consistency property at higher-order overlaps is not isolated explicitly in the published literature.

Conjectured generalization: for an n-fold overlap $\bigcap_{i=1}^n \mathcal{M}^{\sigma_i} \neq \emptyset$, the cocycle chain $u_t^{\sigma_1\sigma_2} \cdot \sigma_t^{\omega^{\sigma_1}}(u_t^{\sigma_2\sigma_3}) \cdot \sigma_t^{\omega^{\sigma_1}}(\sigma_t^{\omega^{\sigma_2}}(u_t^{\sigma_3\sigma_4})) \cdots u_t^{\sigma_{n-1}\sigma_n}$ equals $u_t^{\sigma_1\sigma_n}$ in the canonical gauge. This would give n-fold consistency on the higher-genus overlap; verification is a routine extension of the triple-intersection proof but has not been written down in our sources.

Implication for OSLPLS: higher-order strict super-additivity on n-orbit triples ($n \geq 3$) would follow from the higher-order chain inequality $\sum_{i=1}^{n-1} \Delta S^{\sigma_i \to \sigma_{i+1}} \geq \Delta S^{\sigma_1 \to \sigma_n}$, which is the iterated data-processing inequality.

11.4 Cross-KMS-state thermal time across modular intertwiners

The Tomita–Takesaki modular flow is single-orbit by construction. The Connes cocycle bridges distinct orbits but is itself a single-parameter family of unitaries. A genuine *cross-KMS-state thermal time* that integrates across multiple orbits with full fibration structure (e.g., a thermal-time bundle over the orbit- parameter space) is a natural extension. The Connes–Rovelli 1994 paper [11] does not address this; Paper 49's TICI is a partial step (chain consistency at triple intersections) but does not construct the full fibration.

11.5 Cross-manifold extension (G3 of Paper 45)

The cross-manifold extension $\mathcal{T}_{S^3} \otimes \mathcal{T}_{\mathrm{Hardy}(S^5)}$ is blocked at the NCG-framework level by the Paper 24 [48] §V four-layer Coulomb/HO asymmetry. The modular-Hamiltonian structure of the wedge KMS state (the fourth layer of the asymmetry, added in 2026-05-17) is the most directly relevant to the OSLPLS construction: the SU(3) Wilson on S^5 Bargmann lattice does NOT admit a wedge with integer-spectrum modular Hamiltonian (no spinor sector, no second/first-order distinction, m_l integer-not-half-integer). Hence the cross-manifold OSLPLS construction is structurally blocked by the asymmetry.

This is orthogonal to Paper 49's strict-strong-form Krein–MS–OSLPLS bridge; it is named here as the natural NCG-framework-level question that the OSLPLS construction makes visible.

11.6 Inner-factor calibration data (G4 / W3 question)

The W3 question of Paper 32 [50] §VIII.C (whether calibration data — Yukawa values, Higgs mass, generation count — emerge from a "second packing axiom" or similar structural principle) is orthogonal to the OSLPLS bridge. Paper 49 establishes the operator-algebraic-side strict-strong-form bridge structure; the inner-factor calibration data remains an empirical input.

The Sprint W3 calibration-data investigation (2026-05-08) and its follow-ups returned three clean negatives across independent candidate frameworks (PMNS, lepton mass, mechanical basis); the question remains open without a concrete proposal. OSLPLS does not move this question.

12 Conclusion

Paper 49 closes Q1′ of the GeoVac math.OA arc at the OSLPLS / state level (the metric-level convergence leg B4′ is descoped, as it inherits the Paper 45/46 degeneracy) by extending the K$^+$-weak-form Krein–Mondino–Sämann bridge of Paper 48 [60] to the strict-strong-form on the full $M \neq 0$ Paper 46 [58] Appendix B enlarged Krein substrate. The construction proceeds via three structural moves: (i) replace the synthetic-side target category $\mathbf{LorPLG}_{\mathrm{cov}}$ with the operator-system extension $\mathbf{OSLPLS}_{\mathrm{cov}}$ (containing MS as a faithful sub-category); (ii) identify the bridge functor W^{flip} on objects and morphisms via the Connes–vS contravariant UCP convention; (iii) construct the Connes–Rovelli thermal-time stack across distinct KMS states, with the triple-intersection cocycle identity (TICI, Theorem 6.2) as the stack-consistency property and strict super-additivity (Theorem 6.3) via Datta's max-divergence chain inequality.

The headline structural result is the Bridge Theorem 6.4′-Q1′ (Theorem 7.5), with the substantive new content being the off-orbit strict super-additivity established via the thermal-time stack and the data-processing inequality. The headline physical reading is the *twin paradox as quantum information*: the reverse triangle inequality of synthetic Lorentzian geometry, at the operator-algebraic level, rests on the data-processing inequality of quantum information theory — a substantive new connection between modular theory of operator algebras and synthetic Lorentzian geometry.

The numerical verification panel (§8) confirms all load-bearing predictions at the bit-exact panel cells $(n_{\mathrm{max}}, N_t) \in \{(2,3),(3,5),(4,7)\}$: Lambda joint propinquity bit-exact with Paper 45 reference; Umegaki relative-entropy panel deficits substantially positive (66–81 nats, illustrative; load-bearing D_{max} chain verified 96/96); Riemannian-limit recovery at $N_t = 1$ bit-exact; propagation number $= 2$ at the smallest cell.

Honest scope. The bridge is established at the strict-strong-form OSLPLS level on the full $M \neq 0$ enlarged substrate, at finite cutoff. The closed-form expression for the cocycle entropy production deficit is named open (§11.2). The genuine non-commutative Mondino–Sämann pre-length space concept (Q2′, §11.1) remains the named NCG-research frontier. OSLPLS *contains* MS as a faithful sub-category; it is not a direct synthetic extension of the MS concept.

Place in the math.OA series. Eleventh in the GeoVac math.OA arc (siblings: Papers 29 [49], 32 [50], 38 [51], 39 [52], 40 [53], 42 [54], 43 [55], 44 [56], 45 [57], 46 [58], 47 [59], 48 [60]). Builds on Paper 48 as direct predecessor. Together with Paper 48, completes the Krein–MS bridge program at both K^+-weak-form (Paper 48) and strict-strong-form (Paper 49) rigor.

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